JAIN
DEEMED-TO-BE UNIVERSITY

SCHOOL OF
COMPUTER
SCIENCE AND IT

MSc-CSIT

Semester – 3

Hand Book

AY: 26-27 [ODD] **Batch: 2025-2027**

JAIN

DEPARTMENT OF COMPUTER SCIENCE AND IT

Approved Course Matrix

Program : MSCSIT - MASTER OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Specialization : MSCSITGEN - MSC CSIT GENERAL

For The Batch2025 Semester 3

Total Credits : 22 **No. of Courses In The Semester : 10**

Sr No	Course Type	Basket Set	Course Code	Course Description	Major	Minor	General	Special Fee	Type	Credit	Term No	L-T-P-E	CA Marks		End Sem Exam Theory Marks		End Sem Exam Practical Marks		Marks for QP	Passing Marks
													MAX	MIN	MAX	MIN	MAX	MIN		
1	CC		25MCSIT301	ANALYSIS AND DESIGN OF ALGORITHMS	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
2			25MCSIT302	CLOUD COMPUTING	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
3			25MCSIT303	STATISTICS AND DATA ANALYTICS	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
4			25MCSIT301L *	ANALYSIS AND DESIGN OF ALGORITHMS LAB	Yes	-	-	-	C	1	1	0-0-2-0	100		-	-	-	-	0	40
5			25MCSIT303L *	STATISTICS AND DATA ANALYTICS LAB	Yes	-	-	-	C	1	1	0-0-2-0	100		-	-	-	-	0	40
6	DSE	1	25MCSIT3041	CYBER FORENSICS	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
7		1	25MCSIT3042	NATURAL LANGUAGE PROCESSING	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
8		1	25MCSIT3043	AGILE SOFTWARE DEVELOPMENT APPROACHES	Yes	-	-	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
9	OE	1	25JUOE0911	FINANCIAL LITERACY AND PLANNING	-	-	Yes	-	T	3	1	2-0-0-3	50	-	50	18	-	-	50	40
10		1	21JUOE1408	INTRODUCTION TO PHOTOGRAPHY	-	-	Yes	-	T	3	1	3-0-0-0	50	-	50	18	-	-	50	40
11	SEC		25TDPCLP23 *	TRANSDISCIPLINARY PROJECT CENTRIC LEARNING II	Yes	-	-	-	C		1	0-0-0-9	-	-	-	-	-	-	0	-
12		1	25CSPGET31 *	EMPLOYABILITY SKILL TRAINING 3	Yes	-	-	-	C	2	1	0-0-4-0	100		-	-	-	-	0	40
13		1	25CSPGEP31 *	ENTREPRENEURSHIP SKILL 3	Yes	-	-	-	C	2	1	0-0-4-0	100		-	-	-	-	0	40
14		2	25MCSIT305SI *	SUMMER INTERNSHIP	Yes	-	-	-	C	3	1	0-0-0-9	100		-	-	-	-	0	40

Note: # These are Value Enhancement Course.
+ These Course are applicable Lateral Entry Students Only.
* There is no end semester exam for this Course.

Director Academics

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DEPARTMENT OF COMPUTER SCIENCE AND IT

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Sr No	Course Type	Basket Set	Course Code	Course Description	Major	Minor	General	Special Fee	Type	Credit	Term No	L-T-P-E	CA Marks		End Sem Exam Theory Marks		End Sem Exam Practical Marks		Marks for QP	Passing Marks
													MAX	MIN	MAX	MIN	MAX	MIN		
15	SEC	2	25MCSIT305CP *	CAPSTONE PROJECT	Yes	-	-	-	C	3	1	0-0-0-9	100		-	-	-	-	0	40

Note: # These are Value Enhancement Course.
+ These Course are applicable Lateral Entry Students Only.
* There is no end semester exam for this Course.

Director Academics

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Department of Computer Science & IT

Programme: Master of Science in Computer Science and IT

Time Table

Semester : 3 **AY: 26-27 [ODD]** **Room Number: 812** **Batch: 2025-2027** **Class Teacher: Dr. Haripriya V**

	10.00 – 10.50	10.50 – 11.40	11.45 - 12.35	12.35 - 1.25	1.25-2.10	2.10 - 3.00	3.00 - 3.50	3.50 - 4.40
Mon	ADA-B2-611-MB		ADA - RBS	CF-MB/NLP-GD	LUNCH	CC - BS		
Tue			EST-PR	CF-MB/NLP-GD		SDA-B2-611-MB		
Wed	ADA-B1-714-RBS		SDA - HP	CF-MB/NLP-GD		OE	EST-PR	CC - BS
Thu			SDA - HP	ADA - RBS		OE	SDA-B1-714-HP	
Fri			SDA - HP	CC - BS		OE	ADA - RBS	EST-MB
Sat	Soft Skill	Mentoring	TD-PCL	SI / CP				

Course Code	Course Name	Faculty Name	Open Electives
25MCSIT301	Analysis and Design of Algorithms [ADA]	Prof. Rakshitha B S [RBS]	
25MCSIT302	Cloud Computing [CC]	Dr. Balamurugan S [BS]	
25MCSIT303	Statistics and Data Analytics [SDA]	Dr. Haripriya V [HP]	
25MCSIT3041	Cyber Forensics [CF]	Prof. Meghana Bharajdwaj [MB]	
25MCSIT3042	Natural Language Processing [NLP]	Dr. Ganesh D [GD]	Soft Skill – Prof. Jyothi Shanbagh
25CSPGET31	Employability Skill Training-3 [EST]	Dr. Haripriya V [HP]	
25MCSIT301L	Analysis and Design of Algorithms Lab	Prof. Rakshitha B S [RBS] – Batch - 1 Prof. Meghana Bharajdwaj [MB] – Batch - 2	25JUOE0911- Financial Literacy And Planning
25MCSIT303L	Statistics and Data Analytics Lab	Dr. Haripriya V [HP] – Batch - 1 Prof. Meghana Bharajdwaj [MB] – Batch - 2	21JUOE1408 - Introduction To Photography
25TDPCLP23	Transdisciplinary Project Centric Learning [TD-PCL]	All faculties	
	Mentoring	All faculties	
25MCSIT305SI / 25MCSIT305CP	Summer Internship [SI] / Capstone Project [CP]	Prof. Rakshitha B S [RBS]	

Sd\-	Sd\-	Sd\-	Sd\-
TT Coordinator MSc(CSIT)	Programme Head MSc(CSIT)	HoD CSIT	Director SCSIT

JGI JAIN DEEMED-TO-BE UNIVERSITY | SCHOOL OF COMPUTER SCIENCE AND IT

DEPARTMENT OF COMPUTER SCIENCE AND IT

Programme: MSc (CSIT and FSD) **Semester: 3rd** **AY: 2026-27 (ODD)**

ACADEMIC CALENDAR

Sl. No	Activity Description	Date
1	Registration, Orientation & Commencement of Regular Classes	20-07-2026
2	Workshop -1	07-08-2026
3	Industrial Visit	14-08-2026
4	Technical Workshop Series	28-08-2026
5	Activity-1 Submission	29-08-2026
6	Unit Test-1	03-09-2026 to 05-09-2026
7	TechQuest – An Innovation Challenge by YUKTHI Student Forum	25-09-2026
8	Workshop – 2	05-10-2026
9	e-Sports – Gaming Tournament by YUKTHI Student Forum	02-11-2026
10	Activity-2 Submission	06-11-2026
11	Preparatory Test-2	19-11-2026 to 21-11-2026
12	Last Instruction Day	21-11-2026
13	NOC Submission	26-11-2026
14	Commencement of End Semester Exams	04-12-2026
15	Commencement of 4th Semester Classes	04-01-2027

Continuous Assessment (CA) Scheme and Components: 50 Marks

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Marks
1	Activity 1	Formative	1	15
2	Activity 2	Formative	1	15
3	Unit Test-1	Formative	1	6
4	Preparatory Exam	Formative	1	9
5	Class Participation	Formative	1	5
			Total Marks	**50**

Sd\-
Programme Head
MSc (CSIT and FSD)

Sd\-
HOD

Sd\-
Director

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Course File

Faculty	:	Applied Computing
Department	:	Computer Science and Information Technology
School	:	Computer Science and Information Technology
Program	:	MSc- CSIT and FSD
Program Level	:	
Course Title	:	Analysis and Design of Algorithms
Course Code	:	25MCSIT301
Academic Year	:	2026-27
Semester	:	Third
Semester Start Date	:	
Course Coordinator	:	Rakshitha B S
Course Coordinator email ID	:	Rakshitha.bs@jainuniversity.ac.in
Faculty member Name	:	Rakshitha B S
Faculty member designation	:	Assistant Professor
Faculty member email ID	:	Rakshitha.bs@jainuniversity.ac.in

Teaching Learning and Evaluation Plan (TLEP)

Program Details

Department	:	Computer Science and Information Technology
School	:	Computer Science and Information Technology
Program	:	MSc- CSIT and FSD

Program Educational Objectives:

PEO 1	Domain Knowledge Application and Research orientation: Demonstrate advanced proficiency in applying domain-specific knowledge and conducting research to solve complex problems.
PEO 2	Entrepreneurial Skills and Societal Concerns: Use entrepreneurial skills to identify opportunities, create sustainable ventures, and contribute to economic and social development.
PEO 3	Ethical Values: Uphold ethical values and professional integrity.
PEO 4	Lifelong Learning: Demonstrate a commitment to lifelong learning and continuous professional development.

Program Outcomes (POs) :

PO01	**Computational Knowledge**	Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements
PO02	**Problem Solving / Analytical reasoning / Scientific reasoning**	Design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines with strong analytical and scientific reasoning.
PO03	**Conduct Investigation of complex computing / Research related skills**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO04	**Developing solutions with societal and environmental**	Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs of society.

	concerns	
PO05	**Modern tools / Techniques Usage**	Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts. Select, adapt, and apply appropriate modern tools, techniques, resources to various computing activities, with an understanding of their limitations.
PO06	**Communication Efficacy**	Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
PO07	**Moral and ethical awareness**	Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO08	**Individual and team work towards innovation and entrepreneurship**	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments using innovation to pursue the opportunity to create value and wealth for the betterment of the individual and society at large.

Course Details

Course Title	:	Analysis and Design of Algorithms
Course Code	:	25MCSIT301
Credits	:	03
L-T-P-E	:	3-0-0-0
Notional Learning Hours	:	30
Contact Hours	:	45
Semester	:	Third
Course Category	:	Theory
Contribution to Major/Minor	:	NA
Course Type	:	Theory
CA:ESE	:	50:50
CA Pass Percentage	:	NA
ESE Pass Percentage	:	40%
Aggregate Pass Marks	:	40%

Nature and Orientation of Course

a. Percentage of the course that is oriented towards:

Skills – 70%	:	• Development of algorithmic thinking and problem-solving abilities. • Design and analysis of efficient algorithms. • Application of Divide-and-Conquer, Greedy, Dynamic Programming, Backtracking, and Branch & Bound techniques. • Computational complexity analysis and optimization skills. • Strong programming and analytical competency development.
Entrepreneurship- 05%	:	• Understanding optimization and decision-making techniques useful in developing software products and technology solutions. • Limited direct focus on business creation, innovation management, or entrepreneurial practices.

Employability- 25%	:	• Core subject for software development, data science, AI/ML, and competitive programming roles. • Frequently tested in technical interviews and coding assessments. • Enhances capabilities required for software engineer, algorithm engineer, and research-oriented positions.

b. Percentage of the course that is relevant towards the development needs

Globally- 60%	:	• Algorithm design and analysis are fundamental to global advancements in software engineering, artificial intelligence, cloud computing, cybersecurity, data analytics, and scientific computing. • The concepts taught are universally applicable across industries and research domains worldwide.
Nationally- 20%	:	• Supports the needs of India's IT industry, digital transformation initiatives, fintech sector, e-governance projects, and technology-driven startups. • Enhances competencies required for national-level competitive examinations and technical careers.
Regionally- 10%	:	• Contributes to solving region-specific computational and optimization problems in sectors such as healthcare, education, transportation, agriculture, and manufacturing.
Locally- 10%	:	• Equips students with problem-solving and software development skills that can be applied to local industry requirements, academic projects, startups, and community-based technological solutions.

c. Percentage of the course that is oriented towards

Indian Knowledge Systems - 0%	:	• The syllabus does not explicitly cover Indian Knowledge Systems, traditional computational methods, or indigenous scientific knowledge.
Sustainable Development Goals (SDGs) -10%	:	• Efficient algorithms contribute to sustainable computing through optimized use of computational resources and energy. • Relevant SDGs include: • SDG 4: Quality Education (development of advanced computational skills) • SDG 9: Industry, Innovation and Infrastructure (efficient computational solutions and innovation) • SDG 11: Sustainable Cities and Communities (optimization algorithms for transportation and resource management) • SDG 12: Responsible Consumption and Production (resource-efficient computing)
Emerging Technologies- 40%	:	• Algorithms are the foundation of modern technologies such as: • Artificial Intelligence (AI) • Machine Learning (ML) • Data Science and Big Data Analytics • Robotics • Cloud Computing • Cybersecurity • Optimization and Decision Support Systems • Topics such as Dynamic Programming, Greedy Algorithms, NP-Completeness, and Approximation Algorithms are extensively used in these domains.

Course Objectives

- To provide learners with a strong foundation in algorithm design and analysis. To develop problem-solving skills through the study and application of major algorithmic paradigms including divide and conquer, greedy, dynamic programming, backtracking, and branch and bound techniques for solving computational problems.
- To enable learners to analyze problem complexity, understand computational intractability through NP-completeness concepts, and apply appropriate algorithmic strategies, including approximation methods, to real-world optimization and decision-making problems.

Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BT Level	Attainment Threshold

CO01	Analyze the efficiency of recursive and non-recursive algorithms by applying asymptotic notations and algorithm analysis frameworks to evaluate their time and space complexity for given computational problems under specified input conditions.	**L4**	**60%**
CO02	Apply divide-and-conquer and greedy algorithmic strategies to design and analyze solutions for sorting, searching, graph traversal, and optimization problems by evaluating their correctness and computational efficiency for given problem instances under specified constraints.	**L3**	**60%**
CO03	Analyze dynamic programming techniques to formulate and optimize solutions for combinatorial and graph-based problems by applying the principle of optimality and evaluating computational efficiency for given problem instances under defined constraints.	**L4**	**60%**
CO04	Analyze backtracking and branch-and-bound techniques to solve combinatorial optimization problems by evaluating solution feasibility, correctness, and computational efficiency for given problem instances under specified constraints.	**L4**	**60%**
CO05	Analyze computational complexity concepts to classify problems in P, NP, NP-Complete, and NP-Hard categories by applying polynomial reduction techniques and evaluating approximation strategies for given computational problems under defined constraints.	**L4**	**60%**

CO–PO Mapping Table (3-Strong Correlation 2- Medium Correlation 1- Low Correlation)								
CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	**3**	**3**	**2**	**–**	**2**	**1**	**–**	**1**
CO2	**3**	**3**	**2**	**1**	**2**	**1**	**–**	**2**

CO3	3	3	3	1	2	1	–	2
CO4	3	3	3	1	2	1	–	2
CO5	3	3	3	1	2	2		

Justification for CO–PO Mapping

1. Strong Computational Foundation (PO1):
   All course outcomes require application of core computational knowledge, including algorithm design paradigms, mathematical reasoning, recurrence relations, graph theory, and complexity analysis.
2. Analytical Problem Solving (PO2):
   The course emphasizes systematic problem decomposition, algorithm selection, correctness verification, and efficiency evaluation, directly strengthening analytical and scientific reasoning skills.
3. Research-Oriented Investigation (PO3):
   Students investigate multiple algorithmic approaches, compare computational efficiencies, analyze trade-offs, and derive valid conclusions through theoretical and empirical evaluation.
4. Algorithm Design for Practical Needs (PO4):
   Efficient algorithms developed through this course contribute to solving real-world computational problems with optimized resource utilization, indirectly addressing societal and technological needs.
5. Use of Modern Computational Tools (PO5):
   Implementation, simulation, benchmarking, and performance evaluation of algorithms require the use of modern programming environments and computational tools.
6. Technical Communication Skills (PO6):
   Students are required to express algorithmic logic, complexity proofs, and optimization strategies through reports, documentation, presentations, and structured technical explanations.
7. Professional and Ethical Awareness (PO7):
   Understanding computational limitations and efficiency constraints promotes responsible and ethical development of scalable and resource-aware computing solutions. However we are not evaluating this aspect as part of their activities.
8. Innovation and Creative Problem Solving (PO8):
   Designing optimized solutions using divide-and-conquer, dynamic programming, backtracking, and approximation techniques encourages innovative thinking.
9. Collaborative Learning and Team-Based Analysis (PO8):
   Algorithm implementation exercises, problem-solving sessions, and case

studies promote teamwork and collaborative exploration of computational strategies.

10. Higher-Order Computational Thinking:
The course develops abstraction, optimization, formal reasoning, and complexity classification skills essential for advanced computing research and professional problem-solving, thereby supporting multiple program outcomes comprehensively.

Syllabus

Module No		Hours
1	**Introduction** : Notion of an Algorithm — Fundamentals of Algorithmic Problem Solving — Important Problem Types — Fundamentals of the Analysis of Algorithmic Efficiency –Asymptotic Notations and their properties. Analysis Framework — Empirical analysis — Mathematical analysis for Recursive and Non-recursive algorithms	9
2	**Divide and conquer method, Greedy method:** Divide and conquer methodology – Merge sort – Quick sort – Binary search – Binary tree traversal – Multiplication of large integers – Strassen's matrix multiplication – Greedy method – Prim's algorithm – Kruskal's algorithm – Dijkstra's algorithm..	9
3	**Dynamic programming**: Introduction, The Principle of Optimality, Problem Solving using Dynamic Programming – Calculating the Binomial Coefficient– Warshall's and Floyd' algorithm – Optimal binary search tree – Knapsack problem – Memory functions.	9
4	**Backtracking, Branch and bound** : Backtracking – N-Queens problem –Subset sum problem – Branch and bound – Assignment problem – Knapsack problem – Traveling salesman problem.	9
5	**Introduction to NP-Completeness:** The class P and NP, Polynomial reduction, NP- Completeness Problem, NP-Hard Problems. Travelling Salesman problem, Hamiltonian problem, Approximation algorithms	9

Text Books

1. *Levitin, A. (2012).* Introduction to the design and analysis of algorithms *(3rd ed.). Pearson.* ***ISBN:*** *978-0-13-231681-1.*

Reference Books

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to algorithms* (4th ed.). MIT Press. **ISBN:** 978-0262046305.
2. Horowitz, E., Sahni, S., & Rajasekaran, S. (2008). *Computer algorithms* (2nd ed.). Universities Press. **ISBN:** 978-8173716126.
3. Skiena, S. S. (2020). *The algorithm design manual* (3rd ed.). Springer. **ISBN:** 978-3030542555.
4. Dasgupta, S., Papadimitriou, C. H., & Vazirani, U. V. (2008). *Algorithms*. McGraw-Hill Education. **ISBN:** 978-0073523408.
5. Sedgewick, R., & Wayne, K. (2011). *Algorithms* (4th ed.). Addison-Wesley. **ISBN:** 978-0321573513.

Web Links:

1.	NPTEL – Design and Analysis of Algorithms (Prof. Madhavan Mukund): Complete syllabus-aligned course for Indian university curriculum.
2.	NPTEL Archived Video Lectures (Full Playlist) Includes asymptotic notation, divide-and-conquer, DP, and NP-completeness.
3.	MIT OpenCourseWare – 6.046J Design and Analysis of Algorithms Advanced and mathematically rigorous.
4.	MIT OCW Lecture Video Collection Direct lecture-by-lecture access.
5.	NPTEL Digimat Video Series Structured by lecture topic; useful for semester planning.
6.	NPTEL Current Course Preview Updated course structure and syllabus mapping.
7.	Class Central – NPTEL DAA Course Easy access with overview and learning path.
8.	MIT OpenCourseWare Main DAA Portal Includes notes, assignments, and lecture videos.
9.	Stanford Algorithms Resources (Tim Roughgarden) Excellent for greedy algorithms, graph algorithms, and NP-completeness.
10.	Abdul Bari Algorithms Playlist (YouTube): https://www.youtube.com/@AbdulBari/playlists?utm_source=chatgpt.com Extremely effective for classroom explanation of:

Session Plan

Detailed Session Plan (45 Hours)
MODULE 1: Introduction to Algorithm Design and Analysis (9 Hours)

Module	Session	Topics	Reading / Reference	Pedagogy / Activity Plan
Module 1	Hour 1	Course introduction, notion of an algorithm, characteristics	Levitin Ch.1	Lecture + discussion
Module 1	Hour 2	Fundamentals of algorithmic problem solving	Levitin Ch.2	Problem analysis activity
Module 1	Hour 3	Important problem types: sorting, searching, graph, combinatorial	Cormen Ch.1	Classification exercise
Module 1	Hour 4	Algorithm efficiency and performance metrics	Cormen Ch.3	Worked examples
Module 1	Hour 5	Asymptotic notations: Big-O, Ω, Θ	Cormen Ch.3	Numerical problem solving
Module 1	Hour 6	Properties of asymptotic notations	Levitin	Proof-based discussion
Module 1	Hour 7	Analysis of non-recursive algorithms	Levitin Ch.2	Board exercises
Module 1	Hour 8	Recursive algorithm analysis and recurrences	Cormen Ch.4	Recurrence solving
Module 1	Hour 9	Empirical analysis and performance measurement	Lab/demo	Execution time comparison
MODULE 2: Divide and Conquer and Greedy Method (9 Hours)				
Module 2	Hour 10	Divide and conquer methodology	Levitin Ch.5	Concept lecture
Module 2	Hour 11	Merge Sort and analysis	Cormen Ch.2	Dry-run exercises
Module 2	Hour 12	Quick Sort and analysis	Cormen Ch.7	Coding walkthrough
Module 2	Hour 13	Binary Search and Binary Tree Traversal	Weiss	Tree tracing activity
Module 2	Hour 14	Multiplication of large integers	Cormen	Mathematical derivation
Module 2	Hour 15	Strassen's Matrix Multiplication	Cormen Ch.4	Matrix-based activity
Module 2	Hour 16	Greedy method fundamentals	Levitin Ch.9	Concept mapping
Module 2	Hour 17	Prim's and Kruskal's algorithms	Cormen MST	Graph construction activity

Module 2	Hour 18	Dijkstra's algorithm	Cormen Graph Algorithms	Shortest path simulation
MODULE 3: Dynamic Programming (9 Hours)				
Module 3	Hour 19	Introduction to Dynamic Programming and Principle of Optimality	Levitin Ch.8	Lecture + examples
Module 3	Hour 20	Calculating Binomial Coefficient	Cormen	Table-filling activity
Module 3	Hour 21	Warshall's Algorithm	Levitin	Matrix tracing
Module 3	Hour 22	Floyd's Algorithm	Cormen	Stepwise execution
Module 3	Hour 23	Optimal Binary Search Tree	Cormen	Cost matrix construction
Module 3	Hour 24	0/1 Knapsack problem	Cormen DP	Dynamic table building
Module 3	Hour 25	Memory functions	Levitin	Coding exercise
Module 3	Hour 26	DP implementation practice	Lab session	Hands-on coding
Module 3	Hour 27	Problem-solving session	Worksheet	Student discussion
MODULE 4: Backtracking and Branch & Bound (9 Hours)				
Module 4	Hour 28	Backtracking fundamentals and state-space trees	Levitin Ch.12	Diagram activity
Module 4	Hour 29	N-Queens problem	Levitin	Board simulation
Module 4	Hour 30	Subset Sum problem	Cormen	Recursive tracing
Module 4	Hour 31	Branch and Bound methodology	Levitin	Lecture
Module 4	Hour 32	Assignment Problem	Cormen	Matrix solving
Module 4	Hour 33	Knapsack using Branch & Bound	Cormen	Comparative activity
Module 4	Hour 34	Traveling Salesman Problem	Cormen	State-space exploration
Module 4	Hour 35	Bounding functions and pruning strategies	Instructor notes	Numerical exercises
Module 4	Hour 36	Problem-solving practice	Worksheet	Team-based activity
MODULE 5: NP-Completeness and Approximation Algorithms (9 Hours)				

Module 5	Hour 37	Introduction to computational complexity	Sipser Ch.7	Lecture
Module 5	Hour 38	Class P and NP	Sipser	Concept discussion
Module 5	Hour 39	Polynomial reduction	Cormen	Reduction examples
Module 5	Hour 40	NP-Complete and NP-Hard problems	Sipser	Problem classification
Module 5	Hour 41	Traveling Salesman problem complexity	Cormen	Analytical discussion
Module 5	Hour 42	Hamiltonian problem	Sipser	Graph-based exercise
Module 5	Hour 43	Approximation algorithms	Vazirani	Case examples
Module 5	Hour 44	Case study: Approximate TSP solutions	Vazirani	Comparative activity
Module 5	Hour 45	Course revision and problem-solving session	All references	Review + quiz

Evaluation Plan

Assessment Scheme: IA: UE - 50:50

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	Activity 1	Formative	1	3	3	3	3	3	15

2	Activity 2	Formative	1	3	3	3	3	3	15
3	Unit Test	Formative	1	2	3	-	-	-	05
4	Class Participation	Formative	--	1	1	1	1	1	05
5	Preparatory Exam	Summative	1	2	2	2	2	2	10
6	University Exam	Summative	1	-	-	-	-	-	50
	Total			-	-	-	-	-	**100**

CA: Instructions and Rubrics

A1 – Activity: MOOC Courses for Analysis and Design of Algorithms

Students shall complete minimum of FIVE of the following courses for a total duration of 20 Hrs and submit the certificate, summary report, and applications of the concepts learned.

ADA Topic	LinkedIn Learning Course	Link
Divide and Conquer	Programming Foundations: Algorithms	https://www.linkedin.com/learning/programming-foundations-algorithms
Greedy Method	Programming Foundations: Algorithms	https://www.linkedin.com/learning/programming-foundations-algorithms
Dynamic Programming	Advanced Algorithms	https://www.linkedin.com/learning/advanced-algorithms
Backtracking	Advanced Algorithms	https://www.linkedin.com/learning/advanced-algorithms
Branch and Bound	Advanced Algorithms	https://www.linkedin.com/learning/advanced-algorithms
Graph Algorithms	Data Structures and Algorithms: Graph Algorithms	https://www.linkedin.com/learning/data-structures-and-algorithms-graph-algorithms
String Matching Algorithms	Data Structures and	https://www.linkedin.com/learning/data-structures-and-algorithms-pattern-matching

	Algorithms: Pattern Matching	
Sorting Techniques	Programming Foundations: Algorithms	https://www.linkedin.com/learning/programming-foundations-algorithms
Searching Techniques	Programming Foundations: Algorithms	https://www.linkedin.com/learning/programming-foundations-algorithms
Complexity Analysis	Programming Foundations: Algorithms	https://www.linkedin.com/learning/programming-foundations-algorithms
Optimization Problems	Advanced Algorithms	https://www.linkedin.com/learning/advanced-algorithms
Mathematical Foundations for Algorithms	Programming Foundations: Discrete Mathematics	https://www.linkedin.com/learning/programming-foundations-discrete-mathematics
Trees and Graphs	Data Structures: Trees	https://www.linkedin.com/learning/data-structures-trees
Recursion and Algorithm Design	Python Data Structures and Algorithms	https://www.linkedin.com/learning/python-data-structures-and-algorithms
Algorithmic Problem Solving	Programming Foundations: Beyond the Fundamentals	https://www.linkedin.com/learning/programming-foundations-beyond-the-fundamentals

Rubrics: A1

Criteria	Excellent (3 marks)	Good (2.5 marks)	Satisfactory (2 marks)	Needs Improvement (1.5 mark)
Course Relevance	Course is highly relevant to current CS curriculum, interests, or career goals.	Good relevance with some direct application to	Generally relevant but not strongly connected to learning goals.	Only partially related to CS or personal goals.

		academic or career goals.		
Completion & Certification	Course fully completed with verified certificate and assessments.	Completed with certificate but minor components (e.g. quizzes) missed.	Completed course, but no certificate or incomplete assessments.	Only part of the course completed.
Technical Depth	Demonstrates advanced understanding of complex technical concepts covered in course.	Good understanding with a few technical gaps.	Adequate grasp of most technical content.	Limited understanding of core technical material.
Application of Knowledge	Clearly applied course knowledge in practical/project work or discussions.	Some evidence of applying knowledge meaningfully.	Minimal application; mostly theoretical understanding.	Very limited attempt to apply learning.
Reflection & Reporting	Submitted a thoughtful and well-structured report/reflection on course experience.	Report submitted with good insight and organization.	Basic report with limited reflection.	Superficial or poorly organized reflection.

CA2:

A2 – Activity: Coding Challenge for Algorithm Development

Students shall participate individually or in teams (maximum 2 students) in a Coding Challenge focused on algorithm development and implementation. The challenge may involve problems related to: Searching and Sorting Algorithms, Dynamic Programming, Greedy Algorithms, Graph Algorithms, String Processing, Recursion

and Backtracking, Data Structures and Optimization Techniques, Real-world Computational Problems.

Students are required to:

1. Analyze the given problem statement.
2. Design an appropriate algorithm.
3. Develop and implement the solution using a programming language of their choice.
4. Test the solution with multiple test cases.
5. Optimize the algorithm for efficiency (time and space complexity).

Submit a report containing:

Problem Statement, Algorithm Design, Flowchart/Pseudocode, Source Code, Test Cases and Outputs, Complexity Analysis, Learning Outcomes

Demonstrate the working of the solution during evaluation/viva.

For the subject **Analysis and Design of Algorithms (ADA)**, the problems should explicitly require students to apply algorithm design paradigms such as:

Divide and Conquer, Greedy Method, Dynamic Programming, Backtracking, Branch and Bound, Graph Algorithms, String Matching Algorithms, Sorting and Searching Techniques, Complexity Analysis, Optimization Problems

Sl. No.	Problem Title	ADA Technique(s)
1	Implement and Compare Merge Sort and Quick Sort	Divide and Conquer
2	Find the kth Smallest Element in an Array	Divide and Conquer
3	Closest Pair of Points Problem	Divide and Conquer
4	Fractional Knapsack Problem	Greedy Method
5	Job Sequencing with Deadlines	Greedy Method
6	Minimum Spanning Tree for a Network	Prim's/Kruskal's Algorithm
7	Dijkstra's Shortest Path Algorithm	Greedy Method
8	Huffman Coding for Text Compression	Greedy Method
9	0/1 Knapsack Problem	Dynamic Programming

10	Longest Common Subsequence (LCS)	Dynamic Programming
11	Matrix Chain Multiplication	Dynamic Programming
12	Coin Change Problem	Dynamic Programming
13	Optimal Binary Search Tree	Dynamic Programming
14	N-Queens Problem	Backtracking
15	Sudoku Solver	Backtracking
16	Graph Coloring Problem	Backtracking
17	Hamiltonian Cycle Problem	Backtracking
18	Traveling Salesman Problem	Branch and Bound / Dynamic Programming
19	String Pattern Matching (KMP Algorithm)	String Algorithms
20	Network Routing using Floyd-Warshall Algorithm	Dynamic Programming / Graph Algorithms

Rubrics: A2

Criteria	Excellent (3 Marks)	Good (2 Marks)	Satisfactory (1 Mark)	Poor (0 Marks)
Problem Understanding & Analysis	Clearly identifies requirements and constraints with complete understanding	Good understanding with minor omissions	Partial understanding of the problem	Poor understanding
Algorithm Design & Logic	Innovative, correct, and efficient algorithm design	Correct algorithm with minor inefficiencies	Basic algorithm that solves the problem partially	Incorrect or incomplete algorithm
Code Quality & Implementation	Well-structured, modular, documented, and error-free code	Mostly correct code with minor issues	Functional code with several coding issues	Incomplete or non-functional code
Testing & Optimization	Extensive testing and effective complexity optimization	Adequate testing and some optimization	Limited testing with little optimization	No testing or optimization

Documentation & Viva Voce	Excellent report and confident explanation of design decisions	Good report and satisfactory explanation	Basic report and limited explanation	Poor documentation and unable to explain work

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Course File

Faculty	:	Applied Computing
Department	:	Computer Science and Information Technology
School	:	Computer Science and Information Technology
Program	:	MSc- CSIT
Program Level	:	PG
Course Title	:	Cloud Computing
Course Code	:	25MCSIT302
Academic Year	:	2026-27
Semester	:	Third
Semester Start Date	:	20-07-2026
Course Coordinator	:	Dr.S.Balamurugan
Course Coordinator email ID	:	balamurugan.s@jainuniversity.ac.in
Faculty member Name	:	Dr.S.Balamurugan
Faculty member designation	:	Asst.Professor
Faculty member email ID	:	balamurugan.s@jainuniversity.ac.in

Teaching Learning and Evaluation Plan (TLEP)

Program Details

Department	:	Computer Science and Information Technology
School	:	Computer Science and Information Technology
Program	:	MSc- CSIT and FSD

Program Educational Objectives:

PEO 1 Domain Knowledge Application and Research orientation: Demonstrate advanced proficiency in applying domain-specific knowledge and conducting research to solve complex problems.

PEO 2	Entrepreneurial Skills and Societal Concerns: Use entrepreneurial skills to identify opportunities, create sustainable ventures, and contribute to economic and social development.
PEO 3	Ethical Values: Uphold ethical values and professional integrity.
PEO 4	Lifelong Learning: Demonstrate a commitment to lifelong learning and continuous professional development.

Program Outcomes (POs) :

PO01	**Computational Knowledge**	Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements
PO02	**Problem Solving / Analytical reasoning / Scientific reasoning**	Design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines with strong analytical and scientific reasoning.
PO03	**Conduct Investigation of complex computing / Research related skills**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO04	**Developing solutions with societal and environmental concerns**	Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs of society.
PO05	**Modern tools / Techniques Usage**	Understand and assess societal, environmental, health, safety, legal, and

		cultural issues within local and global contexts. Select, adapt, and apply appropriate modern tools, techniques, resources to various computing activities, with an understanding of their limitations.
PO06	**Communication Efficacy**	Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
PO07	**Moral and ethical awareness**	Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO08	**Individual and team work towards innovation and entrepreneurship**	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments using innovation to pursue the opportunity to create value and wealth for the betterment of the individual and society at large.

Course Details

Course Title	:	Cloud Computing
Course Code	:	25MCSIT302
Credits	:	03
L-T-P-E	:	3-0-0-0
Notional Learning Hours	:	30
Contact Hours	:	45
Semester	:	Third
Course Category	:	Theory
Contribution to Major/Minor	:	NA
Course Type	:	Theory
CA:ESE	:	50:50
CA Pass Percentage	:	NA
ESE Pass Percentage	:	40%
Aggregate Pass Marks	:	40%

Nature and Orientation of Course

a. Percentage of the course that is oriented towards:

Skills – 60%	:	• Understanding cloud computing fundamentals and distributed systems. • Designing and deploying cloud-based applications. • Implementing cloud security mechanisms. • Managing cloud resources and services. • Developing scalable cloud architectures.
Entrepreneurship- 15%	:	• Understanding cloud service business models. • Exploring cloud-based start-up opportunities. • Awareness of SaaS, PaaS and IaaS business ecosystems.
Employability- 25%	:	• Relevant for Cloud Engineer, DevOps Engineer, Cloud Architect, Site Reliability Engineer and Solutions Architect roles. • Builds competencies required in AWS, Microsoft Azure and Google Cloud platforms. • Supports industry certifications and cloud deployment skills.

b. Percentage of the course that is relevant towards the development needs

Globally- 60%	:	• **Cloud computing forms the backbone of modern digital transformation initiatives worldwide.** • Supports global industries such as healthcare, finance, e-commerce, education and manufacturing.
Nationally- 20%	:	• Aligns with Digital India initiatives.

		• Supports growth of cloud-enabled government services and enterprise applications.
Regionally- 10%	:	• Enables development of cloud solutions for regional industries and start-ups. • Facilitates scalable digital services for education and healthcare sectors.
Locally- 10%	:	• Supports local business digitization and cloud adoption. • Provides technical expertise for small and medium enterprises.

c. Percentage of the course that is oriented towards

Indian Knowledge Systems - 10%	:	• Resource sharing and collaborative knowledge management concepts. • Distributed problem-solving approaches reflecting collective knowledge systems. • Digital preservation of cultural and educational resources through cloud technologies.
Sustainable Development Goals (SDGs) -15%	:	**SDG 4 – Quality Education** • Cloud-based learning platforms and educational resources. **SDG 9 – Industry, Innovation and Infrastructure** • Cloud-enabled innovation and digital infrastructure. **SDG 11 – Sustainable Cities and Communities** • Smart city applications and cloud-based governance. **SDG 12 – Responsible Consumption and Production** • Efficient utilization of computational resources through virtualization.
Emerging Technologies- 40%	:	The course is strongly aligned with the following emerging technologies: • Cloud Computing • Edge Computing • Fog Computing • Server less Computing • Artificial Intelligence as a Service (AIaaS) • Internet of Things (IoT) • Big Data Analytics • Containerization and Kubernetes • Multi-cloud Architectures

Course Objectives

- To introduce the principles of parallel and distributed computing that form the foundation of cloud computing environments.
- To provide understanding of cloud service models, deployment models and cloud ecosystem participants.
- To analyse cloud security threats and evaluate security mechanisms such as encryption, hashing, digital signatures and identity management.
- To study cloud management mechanisms including resource administration, monitoring and workload management.
- To design scalable cloud architectures using workload distribution, resource pooling, elasticity and load-balancing principles.

Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BTL	Threshold
CO1	Understand fundamental principles of parallel and distributed computing to understand cloud computing architectures and environments.	L2	60%
CO2	Demonstrate cloud delivery and deployment models to assess the roles and boundaries of cloud computing from provider and consumer perspectives.	L3	60%
CO3	Analyze cloud security threats and cryptographic techniques including encryption, hashing and identity management to enhance cloud security.	L4	60%
CO4	Evaluate cloud management mechanisms such as remote administration and resource management systems to optimize cloud performance.	L5	60%
CO5	Design and implement cloud-based architectures using workload distribution, resource pooling and elastic provisioning for scalable cloud solutions.	L6	60%

Course Outcome	Program Outcomes (PO)							
	PO01	**PO02**	**PO03**	**PO04**	**PO05**	**PO06**	**PO07**	**PO08**
CO01	3	3	3	3	-	-	-	-
CO02	2	2	2	2	2	-	-	-
CO03	2	2	2	2	-	-	-	-
CO04	2	2	2	2	-	-	-	-
CO05	2	2	2	2	-	-	-	-

Justification for CO–PO Mapping

PO1: Strongly supported through cloud computing fundamentals and architecture design.
PO2: Enhanced through analysis of deployment models and security threats.
PO3: Developed through investigation of cloud technologies and architectures.

PO4: Supported through design of scalable cloud solutions.
PO5: Achieved through usage of cloud platforms and management tools.
PO6–PO8: Indirectly addressed through presentations, teamwork and project activities.

Syllabus

Module No		Hours
1	Principles of Parallel and Distributed Computing: Eras of Computing, Parallel vs. Distributed Computing, Elements of Parallel Computing, Elements of Distributed Computing, Technologies for Distributed Computing. Cloud Computing at a Glance, Building Cloud Computing Environments, Computing Platforms and Technologies.	9
2	Fundamental Concepts and Models: Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, The Cloud Provider Perspective, The Cloud Consumer Perspective, Cloud Deployment Models.	9
3	Basic Terms and Concepts: Threat Agents, Cloud Security Threats, Additional Considerations, Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM).	9
4	Network: Broadband Networks and Internet Architecture, Web Technology, Multitenant Technology, Service Technology. Cloud Management Mechanisms: Remote Administration System, Resource Management System	9
5	Architectures: Workload Distribution Architecture, Resource Pooling Architecture, Dynamic / Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture	9

Text Books

1. Cloud Computing Concepts, Technology & Architecture By Thomas Erl, Zaigham
2. Mastering Cloud Computing by Christian Vecchiola, Rajkumar Buyya, and S. Thamarai Selvi,2013,MK. ISBN: 978-0-12-411454-8.

Reference Books (RB):

1. The Basics of Cloud Computing: Understanding the Fundamentals of Cloud Computing in Theory and Practice, Derrick Rountree, Ileana Castrillo, Syngress, 2013.

Web Links:

1. **NPTEL Cloud Computing – IIT Kharagpur (Prof. Soumya Kanti Ghosh)**
   NPTEL Cloud Computing Course

2. **SWAYAM Cloud Computing Course**
   SWAYAM Cloud Computing

3. **Microsoft Learn – Describe Cloud Computing**
   Microsoft Learn Cloud Computing Fundamentals

4. **AWS Educate**
   AWS Educate Official Portal

5. **AWS Skills Builder**
   AWS Training and Certification

6. **Google Cloud Skills Boost**
   Google Cloud Skills Boost

7. **Google Cloud Documentation**
   Google Cloud Documentation

8. **CloudSim Framework**
   CloudSim Research Framework

9. **Cloud Computing Research Repository**
   arXiv Cloud Computing Papers

Session Plan

Session	Module	Topic	Mode of Delivery	Pedagogical Tool / Engagement	Reading Material	CO
1	M1	Course Introduction and Era of Computing	Lecture	Interactive Discussion	TB1 Ch.1	CO1
2	M1	Parallel vs Distributed Computing	Lecture	Think-Pair-Share	TB2 Ch.1	CO1
3	M1	Elements of Parallel Computing	Lecture	Concept Mapping	TB1 Ch.1	CO1
4	M1	Elements of Distributed Computing	Lecture	Group Activity	TB2 Ch.2	CO1
5	M1	Technologies for Distributed Computing	Lecture	Case Study	TB1 Ch.2	CO1
6	M1	Cloud Computing at a Glance	Lecture	Video Demonstration	TB1 Ch.3	CO1
7	M1	Building Cloud Computing Environments	Lecture	Problem Solving	TB1 Ch.3	CO1
8	M1	Computing Platforms and Technologies	Lecture	Discussion	TB2 Ch.2	CO1
9	M1	Review and Quiz	Assessment	Quiz Activity	Notes	CO1
10	M2	Roles and Boundaries	Lecture	Brainstorming	TB1 Ch.4	CO2
11	M2	Cloud Characteristics	Lecture	Concept Mapping	TB1 Ch.4	CO2
12	M2	Cloud Delivery Models	Lecture	Comparative Analysis	TB1 Ch.5	CO2
13	M2	SaaS, PaaS and IaaS	Lecture	Industry Examples	TB1 Ch.5	CO2
14	M2	Cloud Provider Perspective	Lecture	Case Discussion	TB1 Ch.6	CO2
15	M2	Cloud Consumer Perspective	Lecture	Group Discussion	TB1 Ch.6	CO2
16	M2	Public, Private and Community Cloud	Lecture	Concept Illustration	TB1 Ch.7	CO2

17	M2	Hybrid Cloud Deployment	Lecture	Case Analysis	TB1 Ch.7	CO2
18	M2	Module Assessment	Assessment	MCQ Quiz	Notes	CO2
19	M3	Threat Agents	Lecture	Security Scenario Analysis	TB1 Ch.8	CO3
20	M3	Cloud Security Threats	Lecture	Case Study	TB1 Ch.8	CO3
21	M3	Additional Security Considerations	Lecture	Group Discussion	TB1 Ch.8	CO3
22	M3	Encryption Techniques	Lecture	Demonstration	TB1 Ch.9	CO3
23	M3	Hashing Techniques	Lecture	Problem Solving	TB1 Ch.9	CO3
24	M3	Digital Signatures	Lecture	Demonstration	TB1 Ch.9	CO3
25	M3	Public Key Infrastructure	Lecture	Illustration	TB1 Ch.10	CO3
26	M3	Identity and Access Management	Lecture	Case Analysis	TB1 Ch.10	CO3
27	M3	Security Case Study and Review	Seminar	Group Presentation	Notes	CO3
28	M4	Broadband Networks and Internet Architecture	Lecture	Discussion	TB2 Ch.6	CO4
29	M4	Web Technology	Lecture	Demonstration	TB2 Ch.6	CO4
30	M4	Multitenant Technology	Lecture	Case Study	TB1 Ch.11	CO4
31	M4	Service Technology	Lecture	Collaborative Learning	TB1 Ch.11	CO4
32	M4	Cloud Management Mechanisms	Lecture	Concept Mapping	TB1 Ch.12	CO4
33	M4	Remote Administration System	Lecture	Demonstration	TB1 Ch.12	CO4
34	M4	Resource Management System	Lecture	Practical Illustration	TB1 Ch.12	CO4
35	M4	Resource Optimization Techniques	Lecture	Problem-Based Learning	Notes	CO4
36	M4	Module Review and Quiz	Assessment	Quiz	Notes	CO4
37	M5	Workload Distribution Architecture	Lecture	Case Study	TB1 Ch.13	CO5
38	M5	Resource Pooling Architecture	Lecture	Concept Mapping	TB1 Ch.13	CO5
39	M5	Dynamic Scalability Architecture	Lecture	Demonstration	TB1 Ch.14	CO5
40	M5	Elastic Resource Capacity Architecture	Lecture	Scenario Analysis	TB1 Ch.14	CO5
41	M5	Service Load Balancing Architecture	Lecture	Industry Case Study	TB1 Ch.14	CO5
42	M5	Cloud Bursting Architecture	Lecture	Discussion	TB1 Ch.15	CO5
43	M5	Elastic Disk Provisioning Architecture	Lecture	Demonstration	TB1 Ch.15	CO5
44	M5	Redundant Storage Architecture	Lecture	Collaborative Learning	TB1 Ch.15	CO5
45	M5	Course Revision and Problem Solving Session	Review Session	Comprehensive Quiz	All References	CO5

Detailed Session Plan (45 Hours)

Module	Session	Topics	Reading/Reference	Pedagogy / Activity Plan
Module 1: Principles of Parallel and Distributed Computing (9 Hours)				
M1	Hour 1	Course Introduction, Era of Computing	TB1 Ch.1	Lecture + Discussion
M1	Hour 2	Parallel vs Distributed Computing	TB1 Ch.1	Concept Mapping Activity
M1	Hour 3	Elements of Parallel Computing	TB1 Ch.2	Interactive Lecture
M1	Hour 4	Elements of Distributed Computing	TB1 Ch.2	Group Discussion
M1	Hour 5	Technologies for Distributed Computing	TB2 Ch.1	Case Study Analysis
M1	Hour 6	Cloud Computing at a Glance	TB1 Ch.3	Video Demonstration
M1	Hour 7	Building Cloud Computing Environments	TB1 Ch.3	Problem-Based Learning
M1	Hour 8	Computing Platforms and Technologies	TB2 Ch.2	Industry Examples
M1	Hour 9	Module Review and Quiz	Notes	Quiz + Discussion
Module 2: Fundamental Concepts and Models (9 Hours)				
M2	Hour 10	Roles and Boundaries in Cloud Computing	TB1 Ch.4	Lecture
M2	Hour 11	Cloud Characteristics	TB1 Ch.4	Think-Pair-Share
M2	Hour 12	Cloud Delivery Models (SaaS, PaaS, IaaS)	TB1 Ch.5	Comparative Analysis
M2	Hour 13	SaaS Model Applications	TB1 Ch.5	Case Study
M2	Hour 14	PaaS and IaaS Services	TB1 Ch.5	Demonstration
M2	Hour 15	Cloud Provider Perspective	TB1 Ch.6	Group Activity
M2	Hour 16	Cloud Consumer Perspective	TB1 Ch.6	Discussion
M2	Hour 17	Cloud Deployment Models	TB1 Ch.7	Concept Mapping
M2	Hour 18	Public, Private, Hybrid and Community Clouds	TB1 Ch.7	Case Analysis

Module 3: Basic Terms and Concepts (9 Hours)				
M3	Hour 19	Threat Agents and Cloud Security Basics	TB1 Ch.8	Lecture
M3	Hour 20	Cloud Security Threats	TB1 Ch.8	Case Study
M3	Hour 21	Additional Security Considerations	TB1 Ch.8	Discussion
M3	Hour 22	Encryption Techniques	TB1 Ch.9	Demonstration
M3	Hour 23	Symmetric and Asymmetric Encryption	TB1 Ch.9	Problem Solving
M3	Hour 24	Hashing Algorithms	TB1 Ch.9	Hands-on Activity
M3	Hour 25	Digital Signatures	TB1 Ch.10	Demonstration
M3	Hour 26	Public Key Infrastructure (PKI)	TB1 Ch.10	Concept Illustration
M3	Hour 27	Identity and Access Management (IAM)	TB1 Ch.10	Security Analysis Activity
Module 4: Network and Cloud Management Mechanisms (9 Hours)				
M4	Hour 28	Broadband Networks and Internet Architecture	TB2 Ch.4	Lecture
M4	Hour 29	Web Technology Fundamentals	TB2 Ch.4	Interactive Discussion
M4	Hour 30	Multitenant Technology	TB1 Ch.11	Case Study
M4	Hour 31	Service Technology	TB1 Ch.11	Concept Mapping
M4	Hour 32	Cloud Management Mechanisms Overview	TB1 Ch.12	Lecture
M4	Hour 33	Remote Administration System	TB1 Ch.12	Demonstration
M4	Hour 34	Resource Management System	TB1 Ch.12	Practical Illustration
M4	Hour 35	Cloud Monitoring and Resource Allocation	TB1 Ch.12	Problem-Based Learning
M4	Hour 36	Module Review and Quiz	Notes	Quiz + Discussion
Module 5: Cloud Architectures (9 Hours)				

M5	Hour 37	Workload Distribution Architecture	TB1 Ch.13	Lecture
M5	Hour 38	Resource Pooling Architecture	TB1 Ch.13	Concept Mapping
M5	Hour 39	Dynamic Scalability Architecture	TB1 Ch.14	Demonstration
M5	Hour 40	Elastic Resource Capacity Architecture	TB1 Ch.14	Case Study
M5	Hour 41	Service Load Balancing Architecture	TB1 Ch.14	Industry Example Analysis
M5	Hour 42	Cloud Bursting Architecture	TB1 Ch.15	Discussion
M5	Hour 43	Elastic Disk Provisioning Architecture	TB1 Ch.15	Activity-Based Learning
M5	Hour 44	Redundant Storage Architecture	TB1 Ch.15	Collaborative Learning
M5	Hour 45	Course Revision, Case Studies and Problem Solving Session	All References	Review Quiz + Open Discussion

Evaluation Plan

Assessment Scheme: IA: UE - 50:50

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	Activity 1	Formative	1	3	3	3	3	3	15
2	Activity 2	Formative	1	3	3	3	3	3	15
3	Unit Test	Formative	1	2	3	-	-	-	05
4	Class Participation	Formative	--	1	1	1	1	1	05
5	Preparatory Exam	Summative	1	2	2	2	2	2	10
6	University Exam	Summative	1	-	-	-	-	-	50
	Total			-	-	-	-	-	**100**

CA: Instructions and Rubrics

A1 – Activity: MOOC Courses for Cloud Computing

Students shall complete minimum of FIVE of the following courses for a total duration of 20 Hrs and submit the certificate, summary report, and applications of the concepts learned.

Cloud Computing Topic	LinkedIn Learning / Online Course	Link
Cloud Computing Fundamentals	Cloud Computing Foundations	https://www.linkedin.com/learning/cloud-computing-foundations
Cloud Service Models (IaaS, PaaS, SaaS)	Learning Cloud Computing: Core Concepts	https://www.linkedin.com/learning/learning-cloud-computing-core-concepts
Cloud Architecture	Cloud Architecture: Core Concepts	https://www.linkedin.com/learning/cloud-architecture-core-concepts
Cloud Deployment Models	Cloud Computing Foundations: Cloud Models	https://www.linkedin.com/learning/cloud-computing-foundations
AWS Fundamentals	AWS Essential Training for Architects	https://www.linkedin.com/learning/aws-essential-training-for-architects
Microsoft Azure Fundamentals	Microsoft Azure Fundamentals (AZ-900)	https://www.linkedin.com/learning/microsoft-azure-fundamentals
Google Cloud Fundamentals	Google Cloud Platform Essential Training	https://www.linkedin.com/learning/google-cloud-platform-essential-training
Cloud Security	Cloud Security Basics	https://www.linkedin.com/learning/cloud-security-basics
Identity and Access Management (IAM)	Learning Cloud Security	https://www.linkedin.com/learning/learning-cloud-security
Encryption and Cryptography in Cloud	Cybersecurity Foundations	https://www.linkedin.com/learning/cybersecurity-foundations
Virtualization Technologies	Virtualization Foundations	https://www.linkedin.com/learning/virtualization-foundations
Distributed Computing Concepts	Learning Distributed Computing	https://www.linkedin.com/learning/learning-distributed-computing
Resource Management and Scalability	Cloud Computing Concepts	https://www.linkedin.com/learning/cloud-computing-concepts
Load Balancing and Elasticity	AWS Load Balancing and Auto Scaling	https://www.linkedin.com/learning/aws-load-balancing-and-auto-scaling
DevOps and Cloud Automation	DevOps Foundations	https://www.linkedin.com/learning/devops-foundations
Containerization and Docker	Docker Foundations	https://www.linkedin.com/learning/docker-foundations-professional-certificate

	Professional Certificate	
Kubernetes Fundamentals	Learning Kubernetes	https://www.linkedin.com/learning/learning-kubernetes
Edge and Serverless Computing	Serverless Computing Concepts	https://www.linkedin.com/learning/serverless-computing-concepts
Cloud Monitoring and Administration	Cloud Administration Fundamentals	https://www.linkedin.com/learning/cloud-administration-fundamentals
Cloud Solution Design	Cloud Computing for Business	https://www.linkedin.com/learning/cloud-computing-for-business

Rubrics: A1

Criteria	Excellent (3 marks)	Good (2.5 marks)	Satisfactory (2 marks)	Needs Improvement (1.5 mark)
Course Relevance	Course is highly relevant to current CS curriculum, interests, or career goals.	Good relevance with some direct application to academic or career goals.	Generally relevant but not strongly connected to learning goals.	Only partially related to CS or personal goals.
Completion & Certification	Course fully completed with verified certificate and assessments.	Completed with certificate but minor components (e.g. quizzes) missed.	Completed course, but no certificate or incomplete assessments.	Only part of the course completed.
Technical Depth	Demonstrates advanced understanding of complex technical concepts covered in course.	Good understanding with a few technical gaps.	Adequate grasp of most technical content.	Limited understanding of core technical material.
Application of Knowledge	Clearly applied course knowledge in practical/project work or discussions.	Some evidence of applying knowledge meaningfully.	Minimal application; mostly theoretical understanding.	Very limited attempt to apply learning.

Reflection & Reporting	Submitted a thoughtful and well-structured report/reflection on course experience.	Report submitted with good insight and organization.	Basic report with limited reflection.	Superficial or poorly organized reflection.

CA2:

A2 – Activity: Cloud Computing Mini Project

Students shall design, develop, and present a Cloud-Based Solution that demonstrates the practical application of cloud computing concepts such as virtualization, cloud service models, deployment models, security, scalability, resource management, and cloud architectures.

Sl. No.	Project Title
1	Cloud-Based E-Learning Management System
2	Smart Healthcare Monitoring System using Cloud
3	Cloud-Based Inventory Management System
4	Secure File Storage and Sharing System
5	Cloud-Based Attendance Management System
6	IoT-Based Smart Agriculture Monitoring Platform
7	Cloud-Based Library Management System
8	Cloud-Based Customer Relationship Management (CRM) System
9	Real-Time Weather Monitoring using Cloud Services
10	Cloud-Based Online Examination System
11	Cloud-Based Disaster Recovery System
12	Cloud-Based Food Supply Chain Monitoring System

13	Cloud Resource Monitoring Dashboard
14	Cloud-Based Smart Parking System
15	Serverless Application using AWS Lambda/Azure Functions

Rubrics: A2

Criteria	Excellent (3 Marks)	Good (2 Marks)	Satisfactory (1 Mark)	Poor (0 Marks)
Problem Understanding & Analysis	Clearly identifies requirements and constraints with complete understanding	Good understanding with minor omissions	Partial understanding of the problem	Poor understanding
Algorithm Design & Logic	Innovative, correct, and efficient algorithm design	Correct algorithm with minor inefficiencies	Basic algorithm that solves the problem partially	Incorrect or incomplete algorithm
Code Quality & Implementation	Well-structured, modular, documented, and error-free code	Mostly correct code with minor issues	Functional code with several coding issues	Incomplete or non-functional code
Testing & Optimization	Extensive testing and effective complexity optimization	Adequate testing and some optimization	Limited testing with little optimization	No testing or optimization
Documentation & Viva Voce	Excellent report and confident explanation of design decisions	Good report and satisfactory explanation	Basic report and limited explanation	Poor documentation and unable to explain work

Teaching Learning and Evaluation Plan (TLEP)

Faculty	:	Applied Computing
Department	:	Computer Science and Information Technology
School	:	Computer Science and Information Technology
Program	:	MSc- CSIT and FSD
Program Level	:	PG
Course Title	:	Statistics and Data Analytics
Course Code	:	25MCSIT303
Academic Year	:	2026-27
Semester	:	Third
Semester Start Date	:	
Course Coordinator	:	Dr Haripriya V
Course Coordinator email ID	:	v.haripriya@jainuniversity.ac.in
Faculty member Name	:	Dr Haripriya V
Faculty member designation	:	Assistant Professor
Faculty member email ID	:	v.haripriya@jainuniversity.ac.in

Teaching Learning and Evaluation Plan (TLEP)

1. Program Details

Department	:	CS & IT
School	:	School of CS & IT
Program	:	MSc-CS & IT

2. Program Educational Objectives (PEOs)

PEO 1	Domain Knowledge Application and Research orientation: Demonstrate advanced proficiency in applying domain-specific knowledge and conducting research to solve complex problems.
PEO 2	Entrepreneurial Skills and Societal Concerns: Use entrepreneurial skills to identify opportunities, create sustainable ventures, and contribute to economic and social development.
PEO 3	Ethical Values: Uphold ethical values and professional integrity.
PEO 4	Lifelong Learning: Demonstrate a commitment to lifelong learning and continuous professional development.

3. Program Outcomes (POs) and Program Specific Outcomes (PSOs)

PO01	**Computational Knowledge**	Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements
PO02	**Problem Solving / Analytical reasoning / Scientific reasoning**	Design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines with strong analytical and scientific reasoning.
PO03	**Conduct Investigation of complex computing / Research related skills**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO04	**Developing solutions with societal and environmental concerns**	Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs of society.

PO05	**Modern tools / Techniques Usage**	Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts. Select, adapt, and apply appropriate modern tools, techniques, resources to various computing activities, with an understanding of their limitations.
PO06	**Communication Efficacy**	Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
PO07	**Moral and ethical awareness**	Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO08	**Individual and team work towards innovation and entrepreneurship**	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments using innovation to pursue the opportunity to create value and wealth for the betterment of the individual and society at large.

● 4. Course Details

Course Title	:	Statistics and Data Analytics
Course Code	:	25MCSIT302
Credits	:	03
L-T-P-E	:	3-0-0-0
Notional Learning Hours	:	30
Contact Hours	:	45
Semester	:	III
Course Category	:	Core
Contribution to Major/Minor	:	Major
Course Type	:	Theory
CA:ESE	:	50:50
CA Pass Percentage	:	NA
ESE Pass Percentage	:	40%
Aggregate Pass Marks	:	40%

● 5. Nature and Orientation of Course

Percentage of the course oriented towards:

Skills	:	60%	• Across all five modules, students actively develop technical competencies: statistical inference and resampling (Module I), regression, ANOVA, and decision trees (Module II), stream filtering and Bloom filters (Module III), Apriori and

			clustering algorithms (Module IV), and R programming with visualization (Module V). • The two CAs — MOOC completion and a Mini Project involving real dataset analysis — are both skill-execution tasks. • Out of 45 contact hours, the majority involve problem-solving, algorithm application, and programming, making skill development the dominant orientation of this course.
Entrepreneurship	:	10%	• No module is explicitly designed around entrepreneurial thinking. However, the ability to extract patterns from large datasets, build predictive models, and visualize insights forms the analytical backbone of data-driven decision-making in startups and businesses. • This indirect relevance — particularly from Modules IV and V covering predictive analytics and visualization — justifies a conservative 10% rather than zero.
Employability	:	30%	• Industry demands professionals skilled in Big Data, machine learning, and R is well-established. The Mini Project (CA2) directly replicates a standard data analyst workflow — data sourcing, preprocessing, algorithm selection, model evaluation, and reporting. • The MOOC course adds a verifiable credential to the student's profile. Modules II, IV, and V align closely with job roles such as Data Analyst, ML Engineer, and Business Intelligence Analyst, justifying 30% employability orientation.

Percentage of the course relevant towards development needs:

Globally	:	40%	• Big Data, machine learning algorithms (Apriori, K-Means), and R programming are globally adopted technologies. • Frameworks, tools, and datasets used in the course are internationally standardized, and the skills align with global roles in data science and analytics across multinational organizations.
Nationally	:	30%	• India's growing data economy — driven by initiatives like Digital India, NASSCOM's analytics sector, and the rise of Indian tech firms — creates strong national demand for professionals skilled in predictive analytics and statistical modeling. • The course directly prepares students for the Indian IT and data services industry.

Regionally	:	20%	• Southern India, particularly Bengaluru, is a major hub for data analytics, AI startups, and IT companies. • Skills in stream mining, visualization, and R programming are directly applicable to regional employers in this ecosystem, making the course moderately relevant at the regional level.
Locally	:	10%	• Local relevance is limited as data analytics roles are largely concentrated in urban tech centers rather than local or district-level organizations. • However, local businesses and institutions adopting data-driven practices can benefit from graduates with these skills, justifying a small but non-zero local relevance.

Percentage of the course oriented towards:

Indian Knowledge Systems	:	20%
Sustainable Development Goals (SDGs)	:	SDG 9 – Industry, Innovation and Infrastructure
Emerging Technologies (e.g. AI)	:	40%

• 6. Course Objectives

The course aims to explore the fundamental concepts of Big Data Analytics and develop models using various analysis techniques. It equips students with data mining and predictive analysis skills, integrating cryptography for secure decision-making using Java. Additionally, it imparts knowledge on interaction methods, visualization techniques, and Data Analytics with R.

• 7. Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BT Level	Attainment Threshold
CO01	Apply the concepts of Big Data, statistical methods, and analytical tools to address data-intensive problems and support decision-making.	L3	60
CO02	Analyze data using regression models, time series analysis, ANOVA, decision trees, and classification techniques to identify patterns and relationships.	L4	60
CO03	**Examine** stream data processing techniques, including sampling, filtering, Bloom filters, counting algorithms, and stream mining methods, to identify their roles in real-time data analysis.	L4	60
CO04	Evaluate predictive analytics approaches, frequent itemset mining algorithms, and clustering techniques for extracting knowledge from large datasets.	L5	60

CO05	**Assess** and justify the use of data visualization techniques and R programming tools for effective data analysis, interpretation, and communication of actionable insights.	L5	60

• 8. CO: PO Mapping (3 – Strong, 2 – Medium, 1 – Low Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	-	-	-	1	-	
CO2	3	3	3	3	1	1	-	
CO3	3	3	3	3	-	-	-	
CO4	2	3	2	3	-	2	-	1
CO5	2	-	-	2	2	2	2	1
Articulation	2.6	2.5	2.6	2.75	1.5	1.5	1	1

• 9. Syllabus

Module	Topics	Contact Hours
I	INTRODUCTION TO BIG DATA: Introduction to Big Data - Dimensions of Big Data, Challenges of Conventional Systems, Features of Big Data - Security, Compliance, Auditing and Protection, Big Data Characteristics - Analytic Processes and Tools, Analysis vs Reporting, Modern Data Analytic Tools, Statistical Concepts: Sampling Distributions, ReSampling - Statistical Inference - Prediction Error.	9
II	DATA ANALYSIS: Regression Modelling – Multivariate, Support Vector and Kernel Methods; Analysis of Time Series: Linear Systems Analysis, Correlation & Reliability, Regression – Analysis of Variance (ANOVA), Decision Trees – Association, Classification Trees.	9
III	MINING DATA STREAMS: The Stream Data Model - Sampling of Streams, Filtering Streams, Bloom Filter, Counting Distinct Elements - Estimating the number of 1's in a window, Data Stream Mining, Open Source Tools.	9
IV	PREDICTIVE ANALYSIS: Predictive Analytics – Supervised and Unsupervised Learning, Mining Frequent Itemsets - Market Based Model, Apriori Algorithm – Handling large data sets in Main memory, Limited Pass algorithm - Counting frequent itemsets in a stream, Clustering Techniques – Hierarchical, K-Means – Clustering High Dimensional Data.	9
V	VISUALIZATION AND R-PROGRAMMING: Visualization – Purpose of Visualization, Tree and Graph and Time series data visualization techniques, Basic Interaction Techniques. R Fundamentals - Installing, Loading and Running R programs, Accessing Data and R-Script.	9

Reference Books / Text Books:

• TB1: Berthold, M. R., & Hand, D. J. (Eds.). (2007). "Intelligent data analysis: an introduction". Second Edition, ISBN: 3-540-43060-1, Springer Berlin Heidelberg.

• TB2: Ullman, J. D., & Leskovec, J. (2020). "Mining of massive datasets", Third Edition, Cambridge University Press, ISBN: 978-1-108-47634-8.

• RB1: Franks, B. (2012). "Taming the big data tidal wave", ISBN: 978-1-118-20876, John Wiley & Sons.

• RB2: Myatt, G. J. (2007). "Making sense of data: a practical guide to exploratory data analysis and data mining", ISBN-13: 978-0-470-07471-8, John Wiley & Sons.

• RB3: Warden, P. (2011). "Big data glossary", ISBN: 978-1-449-30314-3, O'Reilly Media, Inc.

• OR1: https://beyondcorner.com/hdfs-master-slave-topology/

● 10. Session Plan

Cohort/Section: A	Faculty Member: Haripriya V, Assistant Professor

Innovative Pedagogies Planned: Flipped Classroom, Blended Learning (LinkedIn Learning), Problem-Based Learning

Innovative Pedagogies: Flipped Classroom sessions are planned for topics including Analytic Processes, Prediction Error, Bloom Filter, Estimating 1s in a Window, Handling Large Datasets, and Visualization Techniques. Blended Learning with LinkedIn Learning and Infosys Springboard resources. Duration: Interaction (30 min) | Teacher Input (20 min) | Wrap-Up (10 min).

Session No.	Module No.	Topic	L-T-P-E	Mode of Delivery	Pedagogical Tool / Engagement	Reading Material	CO
1	I	Introduction to Big Data, Dimensions of Big Data	L	Classroom Interactive / PPT	Lecture, Discussion	TB1, RB1	CO1
2	I	Challenges of Conventional Systems	L	Classroom Interactive / PPT	Lecture	TB1	CO1
3	I	Features of Big Data – Security, Compliance, Auditing and Protection	L	Classroom Interactive / PPT	Lecture	TB1	CO1
4	I	Big Data Characteristics	L	Classroom Interactive / PPT	Lecture	TB1	CO1
5	I	Analytic Processes and Tools	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB1, RB1	CO1

6	I	Analysis vs Reporting; Statistical Concepts: Sampling Distributions	L	Classroom Interactive / PPT	Lecture	TB1	CO1
7	I	Resampling – Statistical Inference	L	Classroom Interactive / PPT	Lecture	TB1	CO1
8	I	Prediction Error	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB1	CO1
9	I	Features of Big Data Security, Compliance, Auditing and Protection (Revision)	L	Classroom Interactive / PPT	Lecture	TB1	CO1
10	II	Regression Modeling	L	Classroom Interactive / PPT	Lecture	TB1, RB1	CO2
11	II	Multivariate Regression	L	Classroom Interactive / PPT	Lecture	TB1	CO2
12	II	Support Vector and Kernel Methods	L	Classroom Interactive / PPT	Lecture	TB1	CO2
13	II	Analysis of Time Series: Linear Systems Analysis	L	Classroom Interactive / PPT	Lecture	TB1	CO2
14	II	Correlation & Reliability	L	Classroom Interactive / PPT	Lecture	TB1	CO2
15	II	Regression – Analysis of Variance (ANOVA)	L	Classroom Interactive / PPT	Lecture	TB1	CO2
16	II	Decision Trees	L	Classroom Interactive / PPT	Lecture	TB1	CO2
17	II	Association	L	Classroom Interactive / PPT	Lecture	TB1	CO2
18	II	Classification Trees	L	Classroom Interactive / PPT	Lecture	TB1	CO2
19	III	The Stream Data Model	L	Classroom Interactive / PPT	Lecture	TB1, TB2, RB2	CO3
20	III	Sampling of Streams	L	Classroom Interactive / PPT	Lecture	TB2	CO3

21	III	Filtering Streams	L	Classroom Interactive / PPT	Lecture	TB2	CO3
22	III	Bloom Filter	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB2	CO3
23	III	Counting Distinct Elements	L	Classroom Interactive / PPT	Lecture	TB2	CO3
24	III	Moments of Streams	L	Classroom Interactive / PPT	Lecture	TB2	CO3
25	III	Estimating the number of 1s in a Window	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB2	CO3
26	III	Data Stream Mining	L	Classroom Interactive / PPT	Lecture	TB2	CO3
27	III	Open Source Tools	L	Classroom Interactive / PPT	Lecture	TB2	CO3
28	IV	Predictive Analytics	L	Classroom Interactive / PPT	Lecture	TB2, RB2	CO4
29	IV	Supervised and Unsupervised Learning	L	Classroom Interactive / PPT	Lecture	TB2	CO4
30	IV	Mining Frequent Itemsets, Market Based Model	L	Classroom Interactive / PPT	Lecture	TB2	CO4
31	IV	Apriori Algorithm	L	Classroom Interactive / PPT	Lecture	TB2	CO4
32	IV	Handling Large Datasets in Main Memory	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB2	CO4
33	IV	Limited Pass Algorithm	L	Classroom Interactive / PPT	Lecture	TB2	CO4
34	IV	Counting Frequent Itemsets in a Stream	L	Classroom Interactive / PPT	Lecture	TB2	CO4
35	IV	Clustering Techniques – Hierarchical	L	Classroom Interactive / PPT	Lecture	TB2	CO4

36	IV	K-Means – Clustering High Dimensional Data	L	Classroom Interactive / PPT	Lecture	TB2	CO4
37	V	Visualization – Introduction & Purpose	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB1, TB2	CO5
38	V	Purpose of Visualization	L	Classroom Interactive / PPT	Lecture	TB1	CO5
39	V	Tree, Graph and Time Series Data Visualization	L	Classroom Interactive / PPT	Lecture	TB1	CO5
40	V	Visualization Techniques	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB1	CO5
41	V	Basic Interaction Techniques	L	Classroom Interactive / PPT	Lecture	TB1	CO5
42	V	R Fundamentals	L	Classroom Interactive / PPT	Lecture	TB2	CO5
43	V	Installing, Loading and Running R programs	L	Classroom Interactive / PPT	Lecture	TB2	CO5
44-45	V	Accessing Data and R-Script	L	Classroom Interactive / PPT / Flip Class	Flipped Classroom	TB2	CO5

• 11. Evaluation Plan

Innovative Assessments Planned: Activity 1 (MOOC Course – 15+ hours, with viva); Activity 2 (Mini Project – Real-world dataset with ML analysis)

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	Activity 1	Formative	1	3	3	3	3	3	15
2	Activity 2	Formative	1	3	3	3	3	3	15
3	Unit Test	Formative	1	2	3	-	-	-	05
4	Class Participation	Formative	--	1	1	1	1	1	05
5	Preparatory Exam	Summative	1	2	2	2	2	2	10

6	University Exam	Summative	1	-	-	-	-	-	50
	Total			**-**	**-**	**-**	**-**	**-**	**100**

- **12. CA: Instructions and Rubrics**

CA1 – Activity 1: MOOC Course

Instructions: Students shall complete minimum of FIVE of the following courses for a total duration of 20 Hrs and submit the certificate, summary report, and applications of the concepts learned.

Here are the 10 LinkedIn Learning certifications in a table format:

#	Title of Certification	Direct Link
1	Statistics Foundations 1: The Basics	Click Here
2	Statistics Foundations 2: Probability	Click Here
3	Learning Data Analytics: 1 Foundations	Click Here
4	Learning Data Analytics Part 2: Extending and Applying Core Knowledge	Click Here
5	Predictive Analytics Essential Training: Data Mining	Click Here
6	Data Science Foundations: Data Mining in R	Click Here
7	Data Science Foundations: Data Assessment for Predictive Modeling	Click Here
8	Machine Learning with Python: Foundations	Click Here
9	Python for Data Science and Machine Learning Essential Training Part 1	Click Here
10	R for Data Science: Analysis and Visualization	Click Here

Rubrics – CA1:

Criteria	Excellent (3 marks)	Good (2.5 marks)	Satisfactory (2 marks)	Needs Improvement (1.5 mark)
Course Relevance	Course is highly relevant to current CS curriculum, interests, or career goals.	Good relevance with some direct application to academic or career goals.	Generally relevant but not strongly connected to learning goals.	Only partially related to CS or personal goals.
Completion & Certification	Course fully completed with verified certificate and assessments.	Completed with certificate but minor components (e.g. quizzes) missed.	Completed course, but no certificate or incomplete assessments.	Only part of the course completed.
Technical Depth	Demonstrates advanced understanding of	Good understanding	Adequate grasp of most	Limited understanding of core technical material.

	complex technical concepts covered in course.	with a few technical gaps.	technical content.	
Application of Knowledge	Clearly applied course knowledge in practical/project work or discussions.	Some evidence of applying knowledge meaningfully.	Minimal application; mostly theoretical understanding.	Very limited attempt to apply learning.
Reflection & Reporting	Submitted a thoughtful and well-structured report/reflection on course experience.	Report submitted with good insight and organization.	Basic report with limited reflection.	Superficial or poorly organized reflection.

CA2 – Activity 2: Mini Project

Instructions: Collect a real-world dataset of your choice from any open-source platform. Apply any one suitable machine learning algorithm (classification or regression) to analyze the data and evaluate the model's performance. Submit findings along with the code and a brief report.

Rubrics – CA2:

Criteria	85–100% (Excellent)	65–85% (Very Good)	50–65% (Satisfactory)	40–50% (Limited)	<40% (Poor)
Dataset Selection & Relevance	Dataset is highly relevant, real-world, with clear objective and good source.	Dataset is relevant and from a valid source; objective is somewhat clear.	Dataset is usable but not clearly aligned to the objective.	Dataset is inappropriate or not real-world; lacks clarity.	No valid dataset selected.
Preprocessing & Feature Engineering	Data is thoroughly cleaned, preprocessed, and feature-engineered correctly.	Most preprocessing done correctly; minor improvements needed.	Basic preprocessing done; some key steps missing.	Minimal or incorrect preprocessing; data issues remain.	No preprocessing done.
Algorithm Selection & Implementation	Algorithm is well-justified, suitable, and correctly implemented.	Algorithm is appropriate and mostly implemented correctly.	Algorithm is somewhat appropriate; implementation has issues.	Algorithm is wrongly chosen or not implemented properly.	No algorithm implemented.
Model Evaluation & Visualization	Metrics used correctly; results interpreted well; visuals are clear and relevant.	Evaluation is mostly correct; visualizations are adequate.	Limited evaluation or unclear visuals; interpretation is basic.	Poor evaluation; missing or irrelevant visuals; no interpretation.	No evaluation done.

Report Presentation & Clarity	Report is well-organized, original, with all sections complete and clear.	Report is mostly clear with minor issues; most sections covered.	Report is average in clarity or organization; some sections missing.	Report is poorly structured, unclear, or copied; lacks completeness.	No report submitted.

• 13. Blended Learning Resources

Blended Learning [P2 – Blended Learning with Hands-on]: Interaction (30 min) | Teacher Input (20 min) | Wrap-Up (10 min)

Sl. No.	Link
1	https://www.linkedin.com/learning/statistics-foundations-3-using-data-sets/discover-samples-confidence-intervals-and-hypothesis-testing?u=92695330
2	https://www.linkedin.com/learning/search?keywords=bloom%20filter&u=92695330
3	https://www.linkedin.com/learning/r-for-data-science-analysis-and-visualization/r-for-data-science?u=92695330
4	https://www.linkedin.com/learning/processing-text-with-r-essential-training/the-emergence-of-text-analytics?u=92695330
5	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012828311693893632524 4_shared/overview

• 14. Mapped MOOC Courses

S. No.	Platform	Topic	CO	Link	Duration
MC-1	Coursera	Introduction to R Programming for Data Science	CO5	https://www.coursera.org/learn/introducton-r-programming-data-science/home/module/1	10 hours

• 15. Web Video Links (WVL)

Code	Module	Link
WVL-1	Module 1	Introduction to Big Data – https://www.youtube.com/watch?v=9QxZhapbo0o
WVL-2	Module 2	Regression Modeling – https://www.youtube.com/watch?v=DtOYBxi4AIE
WVL-3	Module 3	Stream Data Model – https://www.youtube.com/watch?v=V6cg39tLnR0
WVL-4	Module 4	Predictive Analytics – https://www.youtube.com/watch?v=DPAsYtUHkkM; Clustering – https://www.youtube.com/watch?v=3M1wUK2zCKY
WVL-5	Module 5	Visualization in R – https://www.youtube.com/watch?v=_WyUme_H2ZQ; Intro to R – https://www.youtube.com/watch?v=_V8eKsto3Ug

• 16. Web Text Links (WTL)

Code	Module	Link
WTL-1	Module 1	http://www.dbta.com/Columns/Big-Data-Notes/The-Fundamentals-of-Big-Data-Analytics-97569.aspx
WTL-2	Module 2	https://careerfoundry.com/en/blog/data-analytics/data-analysis-techniques/; https://www.cs.waikato.ac.nz/~abifet/MOA/StreamMining.pdf
WTL-3	Module 3	https://www.dummies.com/careers/find-a-job/the-4-vs-of-big-data/
WTL-4	Module 4	https://www.informit.com/articles/article.aspx?p=2248639&seqNum=4; https://www.educba.com/data-mining-cluster-analysis/
WTL-5	Module 5	https://towardsdatascience.com/a-guide-to-data-visualisation-in-r-for-beginners-ef6d41a34174

JAIN (DEEMED-TO-BE UNIVERSITY)
BENGALURU, KARNATAKA

Course File

Faculty	**:**	Applied Computing
Department	**:**	School of Computer Science & IT
School	**:**	School of Computer Science & IT
Program	**:**	MSc - CSIT
Program Level	**:**	PG
Course Title	**:**	Cyber Forensics
Course Code	**:**	25MCSIT3041
Academic Year	**:**	2026-2027
Semester	**:**	3
Semester Start Date	**:**	
Course Coordinator	**:**	Meghana Bhardwaj
Course Coordinator email ID	**:**	
Faculty member Name	**:**	Meghana Bhardwaj
Faculty member designation	**:**	
Faculty member email ID	**:**	

Teaching Learning and Evaluation Plan (TLEP)

Program Details

Department	**:**	School of Computer Science & IT
School	**:**	School of Computer Science & IT
Program	**:**	MSc - CSIT

Program Educational Objectives

PEO 1	Domain Knowledge Application and Research orientation: Demonstrate advanced proficiency in applying domain-specific knowledge and conducting research to solve complex problems.
PEO 2	Entrepreneurial Skills and Societal Concerns: Use entrepreneurial skills to identify opportunities, create sustainable ventures, and contribute to economic and social development.
PEO 3	Ethical Values: Uphold ethical values and professional integrity.
PEO 4	Lifelong Learning: Demonstrate a commitment to lifelong learning and continuous professional development.

Program Outcomes (POs)

Code	PO Title	Description
PO01	**Computational Knowledge**	Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements.
PO02	**Problem Solving / Analytical reasoning / Scientific reasoning**	Design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines with strong analytical and scientific reasoning.
PO03	**Conduct Investigation of complex computing / Research related skills**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO04	**Developing solutions with societal and environmental concerns**	Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs of society.
PO05	**Modern tools / Techniques Usage**	Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts. Select, adapt, and apply appropriate modern tools, techniques, resources to

Code	PO Title	Description
		various computing activities, with an understanding of their limitations.
PO06	**Communication Efficacy**	Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
PO07	**Moral and ethical awareness**	Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO08	**Individual and team work towards innovation and entrepreneurship**	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments using innovation to pursue the opportunity to create value and wealth for the betterment of the individual and society at large.

Course Details

Course Title	**:**	Cyber Forensics
Course Code	**:**	25MCSIT3041
Credits	**:**	03
L-T-P-E	**:**	3-0-0-0
Notional Learning Hours	**:**	30
Contact Hours	**:**	45
Semester	**:**	
Course Category	**:**	Theory
Contribution to Major/Minor	**:**	Major
Course Type	**:**	Theory
CA:ESE	**:**	50:50
CA Pass Percentage	**:**	0%
ESE Pass Percentage	**:**	40%
Aggregate Pass Marks	**:**	40%

Nature and Orientation of Course

Percentage of the course that is oriented towards:

Skills	**60%**	• Cyber Forensics is a hands-on investigative discipline. All 5 modules (evidence acquisition, digital evidence handling, analysis/admissibility, cyber law, forensic tools) require practical exposure to forensic tools and procedures. • The MOOC Certification and Case Study activities directly build investigative and tool-usage skills, justifying the highest share.
Entrepreneurship	**5%**	• The Case Study activity requires students to independently plan and execute a simulated forensic investigation, introducing initiative and structured problem-solving — foundational entrepreneurial traits relevant to independent forensic consulting. • The course is not primarily entrepreneurial in orientation, hence the modest share.
Employability	**35%**	• Digital/cyber forensic investigators are in high demand across law enforcement, banking, and corporate security (DFIR analyst, forensic examiner roles). • MOOC certifications from LinkedIn Learning and the case study are resume-relevant, directly mapping to industry job roles in forensic investigation and incident response.

Percentage of the course that is relevant towards the development needs:

Globally	**30%**	• Digital forensics tools, standards (ISO/IEC 27037, ACPO Good Practice Guide) and investigation methods are internationally standardized. • LinkedIn Learning certifications reflect worldwide demand for forensic investigation expertise.
Nationally	**35%**	• India's IT Act 2000 (and amendments), CERT-In directives, and the growing national cybercrime caseload drive strong demand for forensic investigators across law enforcement and BFSI sectors. • The case study activity can be framed around real Indian cybercrime scenarios, reinforcing national relevance.
Regionally	**20%**	• Bengaluru's IT and cybersecurity ecosystem, along with the Karnataka State Cyber Crime units, make forensic skills directly applicable for Jain University graduates.
Locally	**15%**	• The Case Study activity encourages students to examine locally relevant cybercrime incidents and organizational security practices.

Percentage of the course that is oriented towards:

Indian Knowledge Systems	**15%**	• Module 4 (Introduction to Cyber Laws) is grounded in the Indian legal and regulatory context — the Information Technology Act, 2000, its amendments, and Indian cybercrime jurisprudence.
Sustainable Development Goals (SDGs)	**10%**	• The course aligns with SDG 16 (Peace, Justice and Strong Institutions) by equipping students to support lawful investigation, evidentiary integrity, and access to justice in cyber-enabled crime.
Emerging Technologies (e.g. AI)	**10%**	• Module 1 introduces cryptocurrency, blockchain, and the dark web; Module 5 touches on tool-assisted analysis, acknowledging AI-assisted evidence triage and metadata processing as emerging touchpoints. • The limited percentage reflects that emerging technology exposure is a supporting theme rather than the central focus of the course.

Course Objectives

The objective of this course is to enable students to understand the basics of cybercrime, cyber forensics technology, systems and services, and to learn about digital evidence, acquisition, handling, analysis and admissibility, while gaining familiarity with different tools used for cyber forensics acquisition and analysis.

Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BT Level	Attainment Threshold
CO01	Illustrate the fundamentals of cybercrime, cyber forensics, digital evidence and quality control procedures.	**L2**	60

CO	Course Outcome	BT Level	Attainment Threshold
CO02	Demonstrate the process of forensic data acquisition and analysis, and investigate artifacts in different scenarios.	**L3**	60
CO03	Apply the procedure to perform recovery, seizure, analysis and admissibility of digital evidence using legal procedures and standards.	**L3**	60
CO04	Analyze the documentation and presentation of digital forensic evidence from legal and evidentiary perspectives to establish admissibility in judicial proceedings.	**L4**	60
CO05	Evaluate the effectiveness of various forensic tools and procedures used to acquire, recover and analyze digital evidence.	**L5**	60

Note: CO4 has been mapped to Bloom's Level 4 (Analyze) and CO5 to Bloom's Level 5 (Evaluate), and both outcome statements have been reframed with verbs matching these higher-order cognitive levels.

CO: PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	1	2	1	-	2	-
CO2	2	3	2	2	2	-	2	-
CO3	2	3	3	2	2	1	3	-
CO4	1	2	2	2	1	3	3	1
CO5	2	3	3	3	3	1	2	1
Articulation	**1.8**	**2.6**	**2.2**	**2.2**	**1.8**	**1.7**	**2.4**	**1.0**

Syllabus

Module	Topics	Contact Hours
1	**Introduction to Cyber Forensics:** Introduction - Role of ECD and ICT - Types - Classification - Strategies to Prevent Cybercrimes - Cyber War - Cryptocurrency - Blockchain - Ransomware - Deep Web and Dark Web. Steps in Forensic Investigation - Forensic Examination Process - Classification - Incident and Incident Handling - Incident Response Team.	**9**
2	**Digital Evidence:** Types - Evidence Collection Procedure - Sources of Evidence - Operating Systems, Storage Medium, File Systems - Registry - Artifacts - Impediments to Collection - Challenges. Preliminaries - Acquisition and Seizure - Chain of Custody - Collection Form - Acquisition Procedure - Challenges - Handling - Precautions Involved.	**9**
3	**Analysis and Admissibility of Digital Evidence:** Capturing of Forensic Copy - Email Tracking - Role of Forensic Analyst - Electronic Record: Retention - Rules of Admissibility - Categorization - Pre-trial Preparation - Presenting - Summary of Investigation Process.	**9**
4	**Introduction to Cyber Laws:** Need - Cyber Laws and Legal Issues - Minimizing Risk - Initiatives Promoting Cyber Security - Terms and Terminologies - Indian Cyber Laws - International Cyber Laws.	**9**
5	**Forensic Tools:** Types - Drive Imaging and Validation - Integrity Verification - Data Recovery - Registry Analysis - Password Recovery - Network Analysis - Email Analysis - Metadata Processing.	**9**

Text Books

1. Dejey and Murugan, "Cyber Forensics", 2018, Oxford University Press.

Reference Books

1. John R. Vacca, "Computer Forensics: Computer Crime Scene Investigation", 2015, Second Edition, Charles River Media, Inc.

2. B. Nelson, A. Phillips, F. Enfinger, and C. Steuart, "Guide to Computer Forensics and Investigations", 2019, Sixth Edition, CENGAGE.

Session Plan

Cohort/Section	**:**	
Faculty Member	**:**	Meghana Bhardwaj

Innovative Pedagogies Planned - Flipped Classroom, Case Study Method, Hands-on Lab / Tool Demonstrations

Session No.	Module No.	Topic	Mode of Delivery / Engagement	Pedagogical Tool / Activity	Reading Material	CO
1	**1**	Introduction to Cyber Forensics - Role of ECD and ICT	Classroom Interactive / PPT	Activity: A1	TB1 RB1 WVL1 WTL1	**CO1**
2	**1**	Types and Classification of Cybercrime	Classroom Interactive / PPT			**CO1**
3	**1**	Strategies to Prevent Cybercrimes	Classroom Interactive / PPT			**CO1**
4	**1**	Cyber War	Classroom Interactive / PPT / Flip Class			**CO1**
5	**1**	Cryptocurrency and Blockchain	Classroom Interactive / PPT			**CO1**
6	**1**	Ransomware	Classroom Interactive / PPT			**CO1**
7	**1**	Deep Web and Dark Web	Classroom Interactive / PPT / Flip Class			**CO1**
8	**1**	Steps in Forensic Investigation - Forensic Examination Process, Classification	Classroom Interactive / PPT			**CO1**
9	**1**	Incident and Incident Handling - Incident Response Team (Revision)	Classroom Interactive / PPT			**CO1**
10	**2**	Digital Evidence - Types	Classroom Interactive / PPT		TB1 WVL2 WTL2	**CO2**

Session No.	Module No.	Topic	Mode of Delivery / Engagement	Pedagogical Tool / Activity	Reading Material	CO
11	**2**	Evidence Collection Procedure	Classroom Interactive / PPT			**CO2**
12	**2**	Sources of Evidence - Operating Systems, Storage Medium	Classroom Interactive / PPT / Flip Class			**CO2**
13	**2**	File Systems and Registry	Classroom Interactive / PPT			**CO2**
14	**2**	Artifacts	Classroom Interactive / PPT			**CO2**
15	**2**	Impediments to Collection - Challenges	Classroom Interactive / PPT			**CO2**
16	**2**	Preliminaries - Acquisition and Seizure	Classroom Interactive / PPT / Flip Class			**CO2**
17	**2**	Chain of Custody - Collection Form	Classroom Interactive / PPT			**CO2**
18	**2**	Acquisition Procedure - Challenges, Handling, Precautions Involved	Classroom Interactive / PPT			**CO2**
19	**3**	Capturing of Forensic Copy	Classroom Interactive / PPT	Activity: A2	TB1 RB2 WVL3 WTL3	**CO3**
20	**3**	Email Tracking	Classroom Interactive / PPT / Flip Class			**CO3**
21	**3**	Role of Forensic Analyst	Classroom Interactive / PPT			**CO3**
22	**3**	Electronic Record - Retention	Classroom Interactive / PPT			**CO3**

Session No.	Module No.	Topic	Mode of Delivery / Engagement	Pedagogical Tool / Activity	Reading Material	CO
23	**3**	Rules of Admissibility	Classroom Interactive / PPT / Flip Class			**CO3**
24	**3**	Categorization of Evidence	Classroom Interactive / PPT			**CO3**
25	**3**	Pre-trial Preparation	Classroom Interactive / PPT			**CO3**
26	**3**	Presenting Evidence in Court	Classroom Interactive / PPT			**CO3**
27	**3**	Summary of Investigation Process (Revision)	Classroom Interactive / PPT			**CO3**
28	**4**	Need for Cyber Laws	Classroom Interactive / PPT		TB1 WVL4 WTL4	**CO4**
29	**4**	Cyber Laws and Legal Issues	Classroom Interactive / PPT / Flip Class			**CO4**
30	**4**	Minimizing Risk	Classroom Interactive / PPT			**CO4**
31	**4**	Initiatives Promoting Cyber Security	Classroom Interactive / PPT			**CO4**
32-33	**4**	Terms and Terminologies	Classroom Interactive / PPT / Flip Class			**CO4**
34-35	**4**	Indian Cyber Laws (IT Act 2000 and Amendments)	Classroom Interactive / PPT	Activity: A2 (Case Study documentation)		**CO4**
36	**4**	International Cyber Laws	Classroom Interactive / PPT			**CO4**

Session No.	Module No.	Topic	Mode of Delivery / Engagement	Pedagogical Tool / Activity	Reading Material	CO
37	**5**	Forensic Tools - Types	Classroom Interactive / PPT			**CO5**
38	**4**	Drive Imaging and Validation	Classroom Interactive / PPT / Flip Class			**CO5**
39	**4**	Integrity Verification	Classroom Interactive / PPT			**CO5**
40	**4**	Data Recovery	Classroom Interactive / PPT			**CO5**
41	**4**	Registry Analysis	Classroom Interactive / PPT / Flip Class		TB1 WVL5 WTL5	**CO5**
42	**4**	Password Recovery	Classroom Interactive / PPT			**CO5**
43	**4**	Network Analysis	Classroom Interactive / PPT			**CO5**
44	**4**	Email Analysis	Classroom Interactive / PPT			**CO5**
45	**4**	Metadata Processing (Revision)	Classroom Interactive / PPT			**CO5**

Web Video Links (WVL)

Code	Module	Links
WVL-1	Module 1	Introduction to Digital Forensics - Learn the Basics: https://www.youtube.com/watch?v=u2zgEFm5RHQ Digital Forensics Masterclass - Complete Computer Forensics (Course Overview): https://www.youtube.com/watch?v=tezD4_L6Hls
WVL-2	Module 2	Understanding Chain of Custody in Digital Forensics: https://www.youtube.com/watch?v=vimQuaC3RYM
WVL-3	Module 3	Episode 1 - Introduction to Digital Forensics and Investigations | Computer Forensics Explained: https://www.youtube.com/watch?v=3iPMIY5-nWI
WVL-4	Module 4	Cyber Law in India: Understanding India's IT Act 2000 with Amendments 2008-11 (Pavan Duggal): https://www.youtube.com/watch?v=TAz-E06SdBk Information Technology Act, 2000 - Overview Lecture: https://www.youtube.com/watch?v=czDzUP1HclQ
WVL-5	Module 5	Forensic Investigation with FTK Imager & Autopsy GUI: https://www.youtube.com/watch?v=2LqfojEu5zQ FTK Imager - Forensic Acquisition Tool Tutorial: https://www.youtube.com/watch?v=8Hrz5P2D6CA

Web Text Links (WTL)

Code	Module	Links
WTL-1	Module 1	Introduction to Digital Forensics - Startertutorials: https://www.startertutorials.com/blog/introduction-to-digital-forensics.html
WTL-2	Module 2	Digital Forensics Basics - Evidence Collection & Chain of Custody - NetworkersHome: https://www.networkershome.com/fundamentals/cybersecurity/digital-forensics-basics-evidence-collection/
WTL-3	Module 3	Cybercrime Module 6 Key Issues: Digital Evidence Admissibility - UNODC: https://www.unodc.org/cld/en/education/tertiary/cybercrime/module-6/key-issues/digital-evidence-admissibility.html
WTL-4	Module 4	Information Technology Act, 2000 (IT Act) - GeeksforGeeks: https://www.geeksforgeeks.org/ethical-hacking/information-technology-act-2000-india/
WTL-5	Module 5	Digital Forensic Tools: Autopsy & FTK Imager - CliffsNotes: https://www.cliffsnotes.com/study-notes/17994507

Evaluation Plan

Innovative Assessments Planned: MOOC Certification (Activity 1), Case Study (Activity 2)

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	**Activity 1 (MOOC Certification)**	Formative	1	3	3	3	3	3	15
2	**Activity 2 (Case Study)**	Formative	1	3	3	3	3	3	15
3	**Unit Test**	Formative	1	2	3	-	-	-	05
4	**Class Participation**	Formative	--	1	1	1	1	1	05
5	**Preparatory Exam**	Summative	1	2	2	2	2	2	10
6	**University Exam**	Summative	1	-	-	-	-	-	50
	Total	-	-	-	-	-	-		**100**

CA: Instructions and Rubrics

A1 - Activity 1: MOOC Course (LinkedIn Learning Certification)

Students shall complete a minimum of FIVE of the following courses for a total duration of 20 Hrs and submit the certificate, summary report, and applications of the concepts learned, mapped to the relevant module(s) of this course.

LinkedIn Learning Courses for Activity 1:

Sl. No.	Course Title	LinkedIn Learning Link	Relevant Module
1	Cybersecurity Foundations: Computer Forensics	https://www.linkedin.com/learning/cybersecurity-foundations-computer-forensics	1
2	Learning Cyber Incident Response and Digital Forensics	https://www.linkedin.com/learning/learning-cyber-incident-response-and-digital-forensics	1
3	Digital Forensics Essentials	https://www.linkedin.com/learning/digital-forensics-essentials	1
4	Learning Computer Forensics	https://www.linkedin.com/learning/learning-computer-forensics-2	2
5	Operating System Forensics	https://www.linkedin.com/learning/operating-system-forensics	2

Sl. No.	Course Title	LinkedIn Learning Link	Relevant Module
6	Web Forensics: Recovering Digital Evidence	https://www.linkedin.com/learning/web-forensics-recovering-digital-evidence	3
7	Cybersecurity Foundations: Governance, Risk, and Compliance (GRC)	https://www.linkedin.com/learning/cybersecurity-foundations-governance-risk-and-compliance-grc-25657666	4
8	Security Frameworks Fundamentals	https://www.linkedin.com/learning/security-frameworks-fundamentals	4
9	Network Forensics	https://www.linkedin.com/learning/network-forensics-23931106	5
10	Explore a Career in Computer Forensics (Learning Path)	https://www.linkedin.com/learning/paths/explore-a-career-in-computer-forensics	All

Rubrics: A1

Criteria	Excellent (3 marks)	Good (2.5 marks)	Satisfactory (2 marks)	Needs Improvement (1.5 marks)
Course Relevance	Course is highly relevant to current cyber forensics curriculum, interests, or career goals.	Good relevance with some direct application to academic or career goals.	Generally relevant but not strongly connected to learning goals.	Only partially related to cyber forensics or personal goals.
Completion & Certification	Course fully completed with verified certificate and assessments.	Completed with certificate but minor components (e.g. quizzes) missed.	Completed course, but no certificate or incomplete assessments.	Only part of the course completed.
Technical Depth	Demonstrates advanced understanding of complex technical concepts covered in course.	Good understanding with a few technical gaps.	Adequate grasp of most technical content.	Limited understanding of core technical material.
Application of Knowledge	Clearly applied course knowledge in practical/project work or discussions.	Some evidence of applying knowledge meaningfully.	Minimal application; mostly theoretical understanding.	Very limited attempt to apply learning.

Criteria	Excellent (3 marks)	Good (2.5 marks)	Satisfactory (2 marks)	Needs Improvement (1.5 marks)
Reflection & Reporting	Submitted a thoughtful and well-structured report/reflection on course experience.	Report submitted with good insight and organization.	Basic report with limited reflection.	Superficial or poorly organized reflection.

A2 - Activity 2: Case Study

Students, individually or in small groups, shall select a real-world or realistic cybercrime/digital-forensic incident (e.g. a data breach, ransomware attack, financial fraud, or a landmark Indian cyber law case) and prepare a structured case study covering: the nature of the incident, the digital evidence involved, the acquisition and chain-of-custody procedure that would be/was followed, the forensic tools applicable for analysis, and the legal/admissibility perspective under the IT Act, 2000. Students shall submit a written report and deliver a short presentation of their findings.

Suggested Case Study Focus Areas (mapped to modules):

- A ransomware or cyber-extortion incident - evidence identification and incident response (Module 1)
- A digital evidence acquisition and chain-of-custody scenario for a seized device (Module 2)
- An email fraud / phishing investigation - forensic analyst's role and admissibility of evidence (Module 3)
- A cybercrime prosecuted under the Information Technology Act, 2000, e.g. Shreya Singhal v. Union of India, or a bank/financial fraud case (Module 4)
- A forensic tool-based investigation using imaging, recovery, or network analysis tools such as FTK Imager, Autopsy, or Wireshark (Module 5)

Rubrics: A2

Criteria	Poor - Below Expectation	Fair - Needs Improvement	Good - Meets Expectations	Excellent - Exceeds Expectations
Incident Understanding & Scope	Vague/incorrect description of the incident (1 Mark)	Basic description with gaps (2 Marks)	Clear, accurate description of the incident (3 Marks)	Thorough, well-researched description with context and impact analysis (4 Marks)
Forensic Process Mapping (Acquisition, Chain of Custody, Analysis)	Little or no mapping of forensic process (1 Mark)	Partial mapping with errors (2 Marks)	Sound mapping of acquisition, custody and analysis steps (3 Marks)	Comprehensive, accurate mapping aligned with standard procedures (4 Marks)

Criteria	Poor - Below Expectation	Fair - Needs Improvement	Good - Meets Expectations	Excellent - Exceeds Expectations
Legal / Admissibility Analysis	Legal aspects not addressed (1 Mark)	Superficial legal discussion (2 Marks)	Reasonable discussion of admissibility and relevant law (3 Marks)	Insightful analysis linking evidence handling to legal admissibility standards (4 Marks)
Tool / Technique Identification	Tools/techniques not identified or incorrect (1 Mark)	Tools identified with limited justification (2 Marks)	Appropriate tools identified with reasoning (3 Marks)	Well-justified selection of tools with clear applicability (4 Marks)
Report Quality & Presentation	Poorly organized, unclear (1 Mark)	Adequate organization, some clarity issues (2 Marks)	Well-organized, clear report and presentation (3 Marks)	Professional, well-structured report and confident presentation (4 Marks)

Assessment Scheme: IA : UE - 50:50

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	**Activity 1 (MOOC Certification)**	Formative	1	3	3	3	3	3	15
2	**Activity 2 (Case Study)**	Formative	1	3	3	3	3	3	15
3	**Unit Test**	Formative	1	2	3	-	-	-	05
4	**Class Participation**	Formative	--	1	1	1	1	1	05
5	**Preparatory Exam**	Summative	1	2	2	2	2	2	10
6	**University Exam**	Summative	1	-	-	-	-	-	50
	Total	-	-	-	-	-	-		**100**

Prepared by: _____

Verified by: _____

Approved by: _____

`

Course File

Faculty	:	Faculty of Applied Sciences
Department	:	Department of CSIT
School	:	School of CSIT
Program	:	MSc.,
Program Level	:	
Course Title	:	Natural Language Processing
Course Code	:	25MCSIT3042
Academic Year	:	Aug 2026
Semester	:	I
Semester Start Date	:	Dr. Raghavendra R
Course Coordinator	:	
Course Coordinator email ID	:	r.raghavendra@jainuniversity.ac.in
Faculty member Name	:	Dr. Ganesh D
Faculty member designation	:	Professor
Faculty member email ID	:	d.ganesh@jainuniversity.ac.in

Teaching Learning and Evaluation Plan (TLEP)

I) Program Details

Department	:	Department of CSIT
School	:	School of CSIT
Program	:	MSc- CSIT

1. Program Educational Objectives:

PEO 1	**Domain Knowledge Application and Research orientation:** Demonstrate advanced proficiency in applying domain-specific knowledge and conducting research to solve complex problems.
PEO 2	**Entrepreneurial Skills and Societal Concerns:** Use entrepreneurial skills to identify opportunities, create sustainable ventures, and contribute to economic and social development.
PEO 3	**Ethical Values:** Uphold ethical values and professional integrity.
PEO 4	**Lifelong Learning:** Demonstrate a commitment to lifelong learning and continuous professional development.

2. Program Outcomes (POs) and Program Specific Outcomes (PSOs):

PO01	Computational Knowledge	Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements.
PO02	**Problem Solving / Analytical reasoning / Scientific reasoning**	Design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines with strong analytical and scientific reasoning.
PO03	**Conduct Investigation of complex computing / Research related skills**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO04	**Developing solutions with societal and environmental concerns**	Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs of society. Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts.
PO05	**Modern tools / Techniques Usage**	Select, adapt, and apply appropriate modern tools, techniques, resources to various computing activities, with an understanding of their limitations.
PO06	**Communication Efficacy**	Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design

		documentation, make effective presentations, and give and understand clear instructions.
PO07	**Moral and ethical awareness**	Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO08	**Individual and team work towards innovation and entrepreneurship**	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments using innovation to pursue the opportunity to create value and wealth for the betterment of the individual and society at large.

II) Course Details

Course Title	:	**Natural Language Processing**
Course Code	:	**25MCSIT3042**
Credits	:	3
L-T-P-E	:	3-0-0-0
Notional Learning Hours	:	45
Contact Hours	:	45
Semester	:	3
Course Category	:	
Contribution to Major/Minor	:	Major
Course Type	:	Elective
CA:ESE	:	50:50
CA Pass Percentage	:	-
ESE Pass Percentage	:	40%
Aggregate Pass Marks	:	40%

1. Nature and Orientation of Course

a. Percentage of the course that is oriented towards:

Skills 45%	:	The NLP course strongly develops technical and analytical skills required in Artificial Intelligence and language technologies. **Key Skills Developed:** • Text preprocessing (tokenization, stemming, lemmatization) • Language modeling and probability-based analysis • Part-of-Speech (POS) tagging and parsing • Semantic analysis and word sense disambiguation • Implementation using Python-based NLP tools

		Technical Competencies: • Machine learning for text processing • Syntax and semantic structure analysis • Building NLP pipelines for real-world text data **Outcome:** Students gain strong AI and programming skills for text analytics.
Entrepreneurship 15%	**:**	The course enables students to create innovative AI-based language technology solutions and startups. **Entrepreneurial Opportunities:** • AI chatbot development services • Resume screening and HR automation tools • Sentiment analysis tools for businesses • Translation and language processing applications • Content summarization tools for media and education **Startup Potential:** • NLP-based SaaS products • AI-driven customer support systems • Educational AI tools for learning platforms **Outcome:** Encourages innovation and AI-based entrepreneurship.
Employability 40%	**:**	The NLP course is highly aligned with industry job roles in AI, data science, and software development. **Job Roles Supported:** • NLP Engineer • Data Scientist • AI/ML Engineer • Computational Linguist • Software Developer (AI domain) **Industry Applications:** • Chatbots (customer service automation) • Search engines and recommendation systems • Speech recognition systems • Social media analytics and sentiment analysis

		Industry Skills: • Python programming (NLTK, spaCy) • Machine learning models for text • Deep learning for language understanding **Outcome:** Enhances employability in AI, IT, and data-driven industries.

b. Percentage of the course that is relevant towards the development needs

Globally 40%	**:**	NLP is a core technology in the global Artificial Intelligence ecosystem and supports modern digital transformation. **Key Contributions:** • Powers global AI systems like **chatbots, search engines, voice assistants** • Enables multilingual communication and translation systems • Supports data-driven decision-making using text analytics **Global Applications:** • Google Search, ChatGPT, Alexa, Siri • Machine translation (Google Translate) • Sentiment analysis in global social media platforms **Outcome:** Prepares learners for global AI and NLP industry roles.
Nationally 30%	**:**	The course strongly supports India's AI, Digital India, and data-driven governance initiatives. **Key Contributions:** • Enhances AI capabilities for Indian languages (Hindi, Tamil, Kannada, etc.) • Supports **Digital India, AI for All, and Smart Governance** • Helps industries adopt AI-based automation

		Applications in India: • Government chatbot services (UMANG, DigiLocker support systems) • Banking automation and fraud detection • Indian language translation systems • AI-driven education platforms (SWAYAM, online learning tools) **Outcome:** Builds skilled AI workforce for India's growing tech ecosystem.
Regionally 20%	**:**	Supports regional language processing and localized AI solutions. **Key Contributions:** • Development of NLP tools for **regional Indian languages** • Enables regional e-governance communication systems • Improves access to technology in rural and semi-urban areas **Applications:** • Kannada/Tamil/Regional language chatbots • Local government grievance redressal systems • Regional news and sentiment analysis systems **Outcome:** Promotes digital inclusion at regional level.
Locally 10%	**:**	Directly benefits small businesses, education, and local service systems. **Key Contributions:** • AI chatbots for local businesses and institutions • Automated customer support for SMEs • Localized content summarization and translation tools **Applications:** • College and university AI support systems • Local business marketing analytics • Small enterprise customer service automation

		Outcome: Encourages local entrepreneurship and digital adoption.

c. Percentage of the course that is oriented towards

Indian Knowledge Systems 20%	**:**	Natural Language Processing aligns with Indian Knowledge Systems through the study and processing of **language, grammar, and structured knowledge representation**. **Key Connections:** • Ancient Indian grammar system (e.g., **Panini's Ashtadhyayi**) reflects rule-based language processing similar to NLP grammar models • Structured linguistic analysis aligns with Sanskrit-based linguistic traditions • Knowledge representation and semantics relate to Indian philosophical logic systems (Nyaya system) **NLP Linkage:** • Tokenization and parsing resemble grammatical rule structures • Syntax and semantics analysis align with structured language interpretation **Outcome:** Promotes computational understanding of structured language systems rooted in Indian linguistic heritage.
Sustainable Development Goals (SDGs) *(Please specify which SDG) 35%*	**:**	The NLP course strongly contributes to multiple Sustainable Development Goals. **Industry, Innovation and Infrastructure:** • Develops AI-based language systems • Supports innovation in natural language technologies • Strengthens digital infrastructure and intelligent systems

		Quality Education • Enables intelligent tutoring systems and e-learning platforms • Supports multilingual education tools **Decent Work and Economic Growth** • Creates employment in AI, data science, and NLP industries • Encourages entrepreneurship in AI-based products **Reduced Inequalities** • Enables regional and multilingual language processing • Bridges communication gaps using translation systems **Outcome:** Supports inclusive digital transformation and innovation-driven growth.
Emerging Technologies *(E.g. AI)* *45%*	**:**	The NLP course is strongly aligned with cutting-edge AI and computing technologies. **Artificial Intelligence (AI)** • Core foundation of NLP systems • Chatbots, virtual assistants, sentiment analysis **Machine Learning & Deep Learning** • Text classification and prediction models • Neural networks for language understanding **Cloud Computing** • Deployment of NLP models on cloud platforms • Scalable AI services (Google Cloud NLP, AWS AI services) **Speech & Language Technologies** • Speech recognition and voice assistants • Machine translation systems

		Data Science & Big Data Analytics • Text mining from large datasets • Social media and sentiment analysis **Outcome:** Prepares students for next-generation AI and data-driven technologies.

2. Course Objectives

The objectives of this course are:

- ✓ Understand Fundamentals of Language Processing
- ✓ Develop Knowledge of Text Processing Techniques
- ✓ Apply Machine Learning Concepts in NLP
- ✓ Analyze Syntax and Semantics of Language
- ✓ Understand Discourse and Pragmatic Analysis

3. Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BT Level	Attainment Threshold
CO01	Understand the fundamental concepts of Natural Language Processing, including language modeling techniques, grammar-based and statistical models, and basic NLP challenges.	L2	60%
CO02	Apply lexical and morphological analysis techniques such as tokenization, stemming, spelling correction, and n-gram models for processing natural language text.	L3	60%
CO03	Analyze syntactic structures of natural language using context-free grammars, dependency parsing, and part-of-speech tagging techniques.	L3	60%
CO04	Demonstrate semantic understanding of text using formal logic, word sense disambiguation, and semantic role labeling techniques.	L4	60%
CO05	Apply discourse analysis and coreference resolution techniques along with lexical resources for building advanced NLP applications.	L4	60%

4. CO: PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	**-**	**1**	**-**	**-**	**-**	**-**	**1**	**-**
CO2	**1**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
CO3	**2**	**3**	**3**	**3**	**3**	**2**	**2**	**2**

CO4	**3**	**2**	**2**	**2**	**2**	**1**	**2**	**1**
CO5	**1**	**1**	**1**	**1**	**1**	**1**	**2**	**2**
Articulation								

5. Syllabus

Module	Topics	Contact Hours
I	**INTRODUCTION** Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM - Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance	9
II	**WORD LEVEL ANALYSIS** Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging.	9
III	**SYNTACTIC ANALYSIS** Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing.	9
IV	**SEMANTICS AND PRAGMATICS** Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus.	9
V	**DISCOURSE ANALYSIS AND LEXICAL RESOURCES**	9

	Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution.	

Reference Books:

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
2. Steven Bird, Ewan Klein and Edward Loper, —Natural Language Processing with Python‖, First Edition, O_Reilly Media, 2009.
3. Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015.
4. Richard M Reese, —Natural Language Processing with Java‖, O_Reilly Media, 2015.
5. Nitin Indurkhya and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
6. Tanveer Siddiqui, U.S. Tiwary, —Natural Language Processing and Information Retrieval‖, Oxford University Press, 2008.

III) **Session Plan**

Cohort/Section	**MSc – CSIT 3 Sem**
Faculty Member	**Dr. Ganesh D**

Innovative Pedagogies Planned – if Any

1. **Flipped Classroom Approach**

Students are provided with pre-learning resources such as lecture videos and reading materials on topics like **n-grams, POS tagging, parsing, and word sense disambiguation**. Classroom time is used for discussions, problem-solving, and coding demonstrations.

2. **Programming-Based Learning (Python/NLTK-Based Labs)**

Hands-on implementation using Python libraries such as **NLTK, SpaCy, and CoreNLP** for tasks like tokenization, stemming, POS tagging, parsing, and semantic analysis.

3. Problem-Based Learning (PBL)

Students are given real-world NLP problems such as:

- Spam detection
- Sentiment analysis
- Spell correction systems
  to develop algorithmic solutions using NLP concepts.

4. Corpus-Based Learning

Use of real text corpora (news articles, social media data, review datasets) for training and evaluating NLP models such as n-grams, tagging, and classification systems.

5. **Case Study Analysis**

Study of real-world NLP applications like Google Search, Chatbots, Machine Translation systems, and Voice Assistants (Siri/Alexa) to understand practical deployment of NLP techniques.

6. Mini Projects / Experiential Learning

Session Plan

Session no.	Module No.	Topic	L-T-P-E Component	Mode of Delivery	Pedagogical Tool / Engagement	Reading Material	CO
1	I	Introduction: Origins and challenges of NLP	3-0-0-0	PPT	Lecture + Discussion	Jurafsky & Martin (Speech	CO1

2	I	Grammar-based Language Models	3-0-0-0	PPT	Lecture + Examples	Jurafsky & Martin	CO1
3	I	Statistical LM – Regular Expressions	3-0-0-0	PPT	Lecture + Demo	Jurafsky & Martin	CO1
4	I	Finite-State Automata	3-0-0-0	PPT	Lecture + Problem solving	Hopcroft & Ullman	CO1
5	I	English Morphology	3-0-0-0	PPT	Lecture + Case study	Jurafsky & Martin	CO1
6	I	Transducers for lexicon and rules	3-0-0-0	PPT	Lecture + Illustration	Jurafsky & Martin	CO1
7	I	Tokenization techniques	3-0-0-0	PPT	Hands-on demo	NLTK Documentation	CO1
8	I	Spelling error detection & correction	3-0-0-0	PPT	Problem solving	Jurafsky & Martin	CO1
9	I	Minimum Edit Distance	3-0-0-0	PPT	Numerical examples	Jurafsky & Martin	CO1
10	II	Unsmoothed N-grams tagging	3-0-0-0	PPT	Lecture + Coding demo	Jurafsky & Martin	CO2
11	II	Evaluating N-grams	3-0-0-0	PPT	Problem solving	Jurafsky & Martin	CO2
12	II	Smoothing techniques	3-0-0-0	PPT	Lecture + Examples	Jurafsky & Martin	CO2
13	II	Interpolation and Backoff	3-0-0-0	PPT	Case study	Jurafsky & Martin	CO2
14	II	Word Classes	3-0-0-0	PPT	Group discussion	Jurafsky & Martin	CO2
15	II	Part-of-Speech Tagging	3-0-0-0	PPT	Lecture + Demo	Jurafsky & Martin	CO2
16	II	Rule-based tagging	3-0-0-0	PPT	Practice session	NLP Toolkits	CO2
17	II	Stochastic & transformation-based tagging	3-0-0-0	PPT	Lecture + comparison	Jurafsky & Martin	CO2
18	II	Issues in PoS tagging	3-0-0-0	PPT	Discussion	Research papers	CO2
19	III	Context-Free Grammars	3-0-0-0	PPT	Lecture + derivation	Jurafsky & Martin	CO3
20	III	Grammar rules for English	3-0-0-0	PPT	Examples + exercises	Jurafsky & Martin	CO3
21	III	Treebanks	3-0-0-0	PPT	Dataset exploration	Penn Treebank docs	CO3

22	III	Normal forms for grammar	3-0-0-0	PPT	Problem solving	Hopcroft & Ullman	CO3
23	III	Dependency Grammar	3-0-0-0	PPT	Visualization demo	Stanford NLP tools	CO3
24	III	Syntactic Parsing	3-0-0-0	PPT	Parsing tree exercises	Jurafsky & Martin	CO3
25	III	Ambiguity in parsing	3-0-0-0	PPT	Case studies	Research papers	CO3
26	III	Dynamic programming parsing	3-0-0-0	PPT	Algorithm walkthrough	Jurafsky & Martin	CO3
27	III	Shallow parsing	3-0-0-0	PPT	NLP toolkit demo	NLTK / SpaCy docs	CO3
28	IV	Requirements for representation	3-0-0-0	PPT	Lecture + discussion	Jurafsky & Martin	CO4
29	IV	First-Order Logic	3-0-0-0	PPT	Problem solving	Logic in AI texts	CO4
30	IV	Description Logics	3-0-0-0	PPT	Concept mapping	AI textbooks	CO4
31	IV	Semantic attachments & word senses	3-0-0-0	PPT	Examples	Jurafsky & Martin	CO4
32	IV	Relations between senses	3-0-0-0	PPT	Discussion	WordNet resources	CO4
33	IV	Thematic roles	3-0-0-0	PPT	Sentence analysis	Jurafsky & Martin	CO4
34	IV	Selectional restrictions & WSD	3-0-0-0	PPT	Case analysis	WordNet	CO4
35	IV	Supervised WSD	3-0-0-0	PPT	Coding demo	ML NLP papers	CO4
36	IV	Dictionary & Thesaurus	3-0-0-0	PPT	Resource exploration	Oxford / WordNet	CO4
37	V	Introduction to discourse analysis	3-0-0-0	PPT	Lecture	Jurafsky & Martin	CO5
38	V	Lexical resources overview	3-0-0-0	PPT	Resource demo	WordNet	CO5
39	V	Discourse segmentation	3-0-0-0	PPT	Examples	NLP research papers	CO5

40	V	Coherence – Reference phenomena	3-0-0-0	PPT	Sentence linking	Jurafsky & Martin	CO5
41	V	Coherence – Reference phenomena (cont.)	3-0-0-0	PPT	Practice	NLP datasets	CO5
42	V	Anaphora resolution (Hobbs algorithm)	3-0-0-0	PPT	Algorithm walkthrough	AI textbooks	CO5
43	V	Centering algorithm for anaphora	3-0-0-0	PPT	Case study	Research papers	CO5
44	V	Coreference resolution	3-0-0-0	PPT	NLP toolkit demo	SpaCy / AllenNLP	CO5
45	V	Advanced coreference resolution	3-0-0-0	PPT	Project discussion	Recent NLP	CO5

Evaluation Plan

Assessment Scheme: IA: UE - 50:50

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	CO1	CO2	CO3	CO4	CO5	Total
1	Activity 1	Formative	1	3	3	3	3	3	15
2	Activity 2	Formative	1	3	3	3	3	3	15
3	Unit Test	Formative	1	2	3	-	-	-	05
4	Class Participation	Formative	--	1	1	1	1	1	05
5	Preparatory Exam	Summative	1	2	2	2	2	2	10
6	University Exam	Summative	1	-	-	-	-	-	50
	Total			-	-	-	-	-	**100**

Activity 1: Problem Solving / Case Study / Assignment (as applicable)

S.No	Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Understanding of Concept	Complete and clear understanding of topic	Minor gaps in understanding	Basic understanding only	Poor or incorrect understanding	4
2	Technical Accuracy	Fully correct solution with proper steps	Minor errors in approach	Some errors in methodology	Incorrect solution	4

S.No	Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
3	Application of Knowledge	Excellent real-world application	Good application	Limited application	No application	4
4	Problem Solving Approach	Systematic, logical, well-structured	Mostly logical	Partially structured	Unstructured	4
5	Presentation & Clarity	Very clear, well-organized, neat	Clear but minor issues	Average presentation	Poor presentation	4

Rubrics: A2

Activity 2: Advanced Problem Solving / Practical Implementation / Case Study

S.No	Criteria	Excellent	Good	Satisfactory	Poor
1	**Conceptual Understanding**	Deep understanding with clear explanation of concepts	Good understanding with minor gaps	Basic understanding	Incorrect or unclear understanding
2	**Implementation / Solution Accuracy**	Fully correct implementation with no errors	Minor logical/technical errors	Partially correct solution	Incorrect or incomplete solution
3	**Application to Real-World Problem**	Strong real-world relevance and insight	Good application	Limited relevance	No application
4	**Analytical Thinking**	Excellent reasoning and structured approach	Good analysis	Basic reasoning	No logical analysis
5	**Output Quality & Documentation**	Well-documented, neat, and complete output	Mostly clear documentation	Average presentation	Poor or missing documentation

JU/JR/SO-05/23-1/2025/510/0.01 Date: 23 December, 2025

REVISED NOTIFICATION

List of General and University Holidays for the year:- 2026

JAIN (Deemed-to-be University), for all **departments** under its ambit, has declared the following General Holidays for the **calendar year 2026**.

Sl. No	Date	Day	Name of the Festival
1.	15.01.2026	Thursday	Uttarayana Punyakala, Makara Sankranti
2.	26.01.2026	Monday	Republic Day
3.	19.03.2026	Thursday	Chandramana Ugadi
4.	21.03.2026	Saturday	Khutub-E-Ramzan
5.	31.03.2026	Tuesday	Mahaveera Jayanthi
6.	03.04.2026	Friday	Good Friday
7.	14.04.2026	Tuesday	Dr. B.R. Ambedkar Jayanthi
8.	20.04.2026	Monday	Basava Jayanthi, Akshaya Tritiya
9.	01.05.2026	Friday	Labour Day / May Day
10.	28.05.2026	Thursday	Bakrid
11.	26.06.2026	Friday	10th Day / Last Day of Muharram (Ashura)
12.	15.08.2026	Saturday	Independence Day
13.	26.08.2026	Wednesday	Eid - Milad
14.	14.09.2026	Monday	Swarnagowri Vratam/ Varasiddhi Vinayaka Vrata
15.	02.10.2026	Friday	Gandhi Jayanthi
16.	10.10.2026	Saturday	Mahalaya Amavasye
17.	20.10.2026	Tuesday	Maha Navami, Ayudhapooja
18.	21.10.2026	Wednesday	Vijayadashami
19.	10.11.2026	Tuesday	Balipadyami, Deepavali
20.	27.11.2026	Friday	Kanakadasa Jayanthi
21.	25.12.2026	Friday	Christmas

The Restricted Holiday announced by the Government and granted as holiday by JAIN (Deemed-to-be University) for the year 2026

Sl. No	Date	Day	Name of the Festival
1.	21.08.2026	Friday	Varamahalakshmi Vrata

Special Holiday Declared by the University Following a Government Holiday

Sl. No.	Date	Day
22.		
23.	20.03.2026	Friday
24.	04.04.2026	Saturday
25.	02.05.2026	Saturday
26.	27.06.2026	Saturday
27.	22.08.2026	Saturday
28.	19.10.2026	Monday
29.	09.11.2026	Monday

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University-Declared Saturday Holiday (As per Approved Academic Calendar Adjustments)

Sl. No	Date	Day
30.	24.01.2026	4th Saturday
31.	21.02.2026	3rd Saturday
32.	04.04.2026	1st Saturday
33.	02.05.2026	1st Saturday
34.	27.06.2026	4th Saturday
35.	18.07.2026	3rd Saturday
36.	22.08.2026	4th Saturday
37.	19.09.2026	3rd Saturday
38.	19.12.2026	3rd Saturday

Note:

- On **17th October,2026**, classes and office work will follow the **same timetable as 19th October, 2026.**
- On **21st November,2026** classes and office work will follow the **same timetable as 9th November, 2026.**

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